

An attempt to connect RTC and SD card to FM-7 (SD card version, part 4: Finally complete!)

 flexonsbd.blogspot.com/2023/04/fm-7rtcscsd_25.html

Tuesday, April 25, 2023

An attempt to connect an RTC and SD card to the FM-7 (SD card version, part 4: Finally complete!)

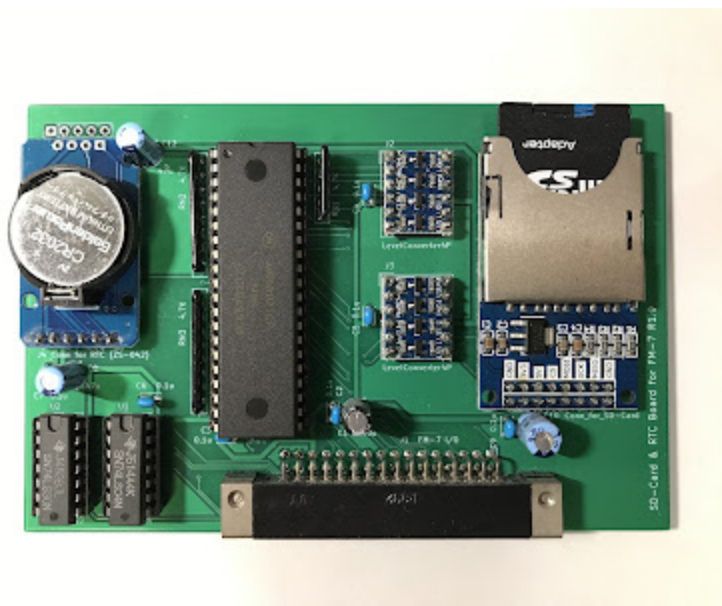
I was finally able to load a BASIC program that had been saved in ASCII format.

[Added May 19th] A revised version has been uploaded as "Part 5," so the program upload mentioned in this article has been removed. If needed, please download from "Part 5."

This was the last remaining hurdle, but it has finally been resolved. Although there are still a few limitations, I would like to consider this complete.

To summarize, I've included the images and circuit diagram again.

The one on the left in the center is the 68B21 (PIA), to its right is the level conversion module (one is a spare), the leftmost is the ZS-042 (RTC module), and the rightmost is the SD card module (with 5V-3.3V level conversion).

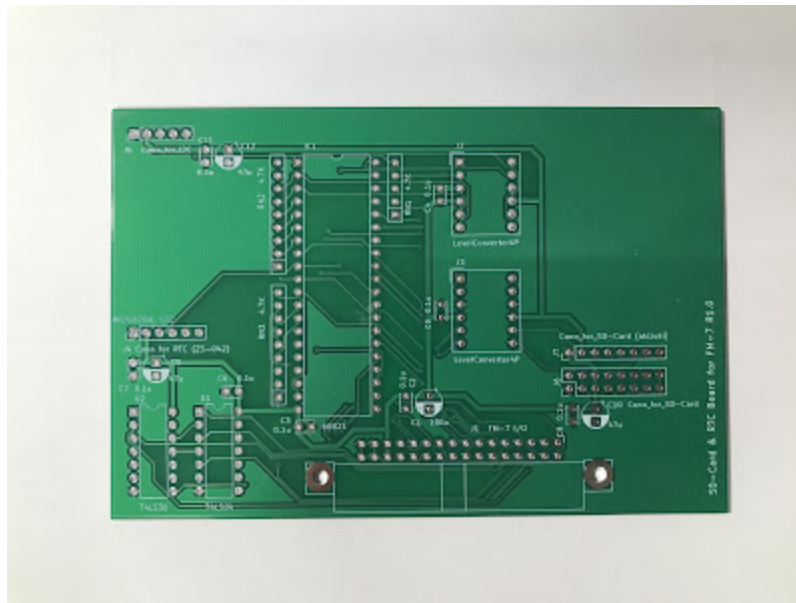


RTC_SD card

The 5-pin connector in the upper left (connector not yet installed) is for I2C expansion. It is intended for connecting I2C LCDs and other devices.

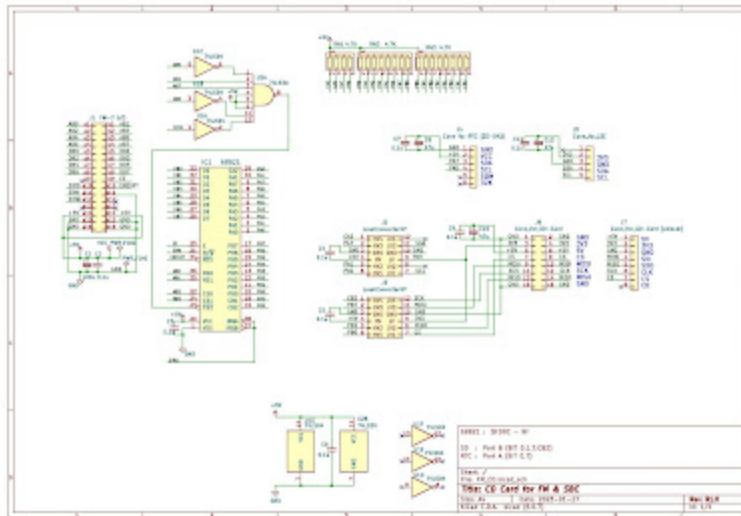
In addition to the SD card module shown in the image, we also have a pattern for one manufactured by Akizuki Denshi.

(Unusually for me, the circuit board worked perfectly on the first try...)



RTC_SD board

I'm using one 68B21 chip to control the RTC module via I2C on port A and the SD card via SPI on port B.



Circuit diagram

I struggled to load the ASCII-saved BASIC program until the very end.

To read a BASIC program file saved in ASCII format, the file must be converted line by line into an intermediate language format and saved in a designated area. For this purpose, F-BASIC provides a "line-by-line translation routine."

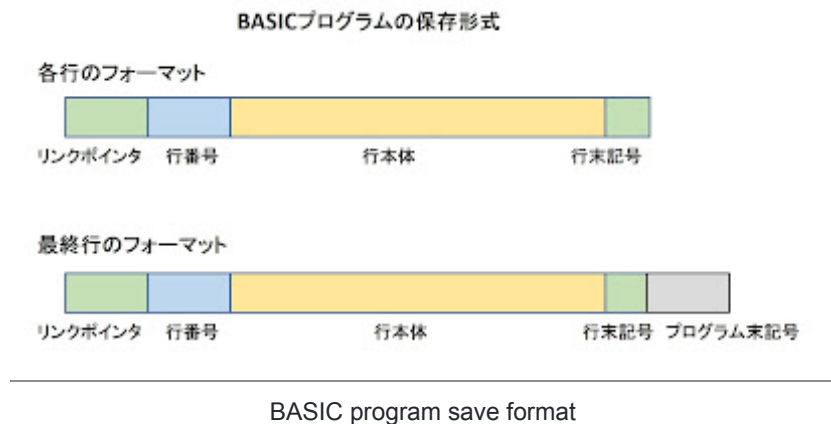
However, I didn't really understand how to use that routine, so initially, I was calling the "line translation routine" every time I read a line from the buffer after reading a file from the SD card, and converting it to an intermediate language format.

The results seemed reasonably correct, and I was even able to execute programs that were only a few lines long after loading them.

However, inexplicable phenomena occur, such as certain commands being mistranslated, or the converted line numbers becoming completely nonsensical values if the line numbers are 100, 110, etc.

While loading various program files, I discovered that the problem stemmed from the fact that the converted file size sometimes became 1 byte instead of 2 bytes depending on the line number value.

In the intermediate language format (2 bytes for the link pointer, 2 bytes for the line number, line body, and \$00 for the line ending), the converted line number must always be 2 bytes.



Therefore, I realized that it seemed necessary to read the line numbers individually and convert them to two bytes, so I changed the code to use the "line number reading routine" provided in F-BASIC. Even then, the conversion didn't work properly at first, but by adding the necessary preprocessing steps to execute this routine, it eventually converted correctly. In the end, the procedure is

- (1) Read the line number portion using the "line number reading routine" and convert it to 2 bytes.
- (2) The line body, excluding the line number, is converted into an intermediate language format using the "single-line translation routine".
- (3) Leave two bytes of space for the link pointer in the memory storage area, then write the line number, followed by the line body converted to intermediate language format, and finally \$00 as the line ending symbol.
- (4) Write the address of the position following the line-ending symbol to the link pointer position at the beginning of the line.

Repeat steps (3) and (4) as long as there are rows.

- (5) Write a 2-byte \$00 as the program end symbol at the position immediately following the end of the last line, and write the address of the beginning of that position to \$35,36.

(I've written quite a lot of details, but please forgive me, as I did so in the hope that it might be of some help to anyone considering a similar endeavor.)

This allows us to read BASIC program files saved in ASCII format. Therefore, the functionality now includes reading BASIC files (binary and ASCII saved) and machine code files stored on the SD card, and writing BASIC files (binary and machine code files) to the SD card, thus achieving our initial objective.

Other necessary peripheral programs

When actually using the system, a timestamp is required when saving files, so you need to read the date and time from the RTC beforehand and set it in F-BASIC.

Since doing this manually is tedious, I decided to set it up by automatically executing the STARTUP file when F-BASIC starts. The two programs I prepared are as follows:

Furthermore, since the main body of the SD card processing program occupies \$6000 or more, machine code files in overlapping areas cannot be read directly. To address this, an offset function of -\$2000 bytes is available. Consequently, a transfer program is required to return the files to their original positions after reading, and a BASIC program for this purpose has also been provided.

SDRWFM SD card read/write program for FM-7

Main unit: \$6000–\$71D1, Buffer: \$5E00–\$5FFF

- STARTUP BASIC program: Calls and executes the machine code routine RTC_SET.
- RTC_SET machine code routine: Reads the RTC and sets the date and time in F-BASIC.
- SDTRS BASIC program: Move the loaded machine code file up to \$2000.

To summarize the features that were implemented:

1. It supports both SD cards and SDHC cards.
2. BASIC files, machine code files, and data files written to an SD card on Windows can be read by the FM-7.
3. BASIC files and machine code files can be written to an SD card using the FM-7 and read on Windows.
4. Commands equivalent to F-BASIC's FILES, LOAD, LOADM, SAVE, SAVEM, and KILL have been implemented.
5. As an optional feature, there is a function to display sectors such as FAT and DIR on the SD card.

Its greatest feature is that it can read and write data saved in the standard Windows format, rather than a special proprietary format.

The constraints are

Since this is intended for use with F-BASIC, sections 1, 2, 3, and 4 below are tailored to that.

1. It does not support large files. (Probably less than approximately 20KB)

2. Directories are not supported.

3. As with F-BASIC, file names do not have extensions.

(It's okay if it's there, but I haven't checked, so I can't distinguish between it and not being there.)

4. BASIC files and machine code files must be saved in the F-BASIC disk format.

5. Lowercase letters cannot be used in filenames. (Long filenames are not supported.)

6. Since the F-BASIC system's date and time are used for the timestamp when saving files, it is necessary to set the correct time beforehand.

(The settings can be read from the RTC during STARTUP.)

7. Only SDHC cards up to 16GB have been confirmed to work. (For some reason, my 32GB card won't work...)

Precautions for use

1. Before starting this program, you must execute CLEAR,&H5DFF.

2. Since the file size is limited to one cluster (64 sectors) of the SD card, the upper limit is $64 \times 512 = 32768$ bytes, and anything exceeding this will be cut off.

(In reality, due to memory area limitations, the upper limit is much lower, probably less than 20KB.)

3. The file format must conform to the F-BASIC standard.

- F-BASIC file (ASCII save): Starts with a newline character (\$0D,\$0A).

- F-BASIC files (binary saves): Starts with \$FF(\$FE),\$FF,\$FF.

- Machine code file: It has a 5-byte header at the beginning (\$00, 2 bytes for size, 2 bytes for starting address) and a 5-byte footer at the end (\$FF, \$00, \$00, 2 bytes for execution address). (In practice, the footer is not checked.)

If none of the above three conditions apply, the data will be loaded as a data file starting from \$2000.

[Caution] Even if it is not a machine code file, if it starts with \$00, it will be mistakenly recognized as a machine code file.

I think there are still some parts that need fixing, but for now, I'll show the program as it is.

```

1      0001      *
2      0002      *   SDRWFM.TXT
3      0003      *
4      0004      *   SD Read/Write for FM-7
5      0005      *
6      0006      *       use RTC & SD Card
7      0007      *
8      0008      *       6821(PIA) Port A -- RTC (I2C Mode)
9      0009      *
10     0010      *           SBC           RTC           Ext IO
11     0011      *   CA2  2  --           --  HV4 LV4 --  I01
12     0012      *   PA0  3  --  SCL  --  HV1 LV1 --  I04
13     0013      *   PA1  4  --           --  HV2 LV2 --  I03
14     0014      *   PA7 10  --  SDA  --  HV3 LV3 --  I02
15     0015      *   GND    --           --           --  GND
16     0016      *           --           --           --  3V3
17     0017      *
18     0018      *
19     0019      *       6821(PIA) Port B -- SD (SPI Mode)
20     0020      *
twenty 0021      *           SBC           SD
one    0022      *   CB2 12  --  HV4 LV4 --  SCK
twenty 0023      *   PB0 13  --  HV1 LV1 --  CS
two    0024      *   PB1 14  --  HV2 LV2 --  MISO
twenty 0025      *   PB7 20  --  HV3 LV3 --  MOSI
three  0026      *   GND 31  --           --  GND
twenty 0027      *   VCC 33  --           --  VCC
four   0028      *
twenty 0029      *       cf. all data line use level converter
five   0030      *
26     0031      *       PB  output mode
27     0032      *       CB2 1bit output
28     0033      *
29     0034      *       READ/WRITE BUFFER  $5E00 - 5FFF
30     0035      *
31     0036      *       Fix date routine          '2023.04.12
32     0037      *       Implement write cmd        '2023.04.07
33     0038      *       Add command                '2023.03.14

```

34	0039		*	Change use Port B	'2023.03.07	
35	0040		*	Original version	'2022.06.11	
36	0041		*	Copyright by Y.Yamamoto		
37	0042		*			
38	0043	6000		STTADR EQU	\$6000	
39	0044					
40	0045	FBB8		INCHA EQU	\$FBB8	for FM-7
41	0046	FBBB		OUTCHA EQU	\$FBBB	for FM-7
42	0047	FBF7		CSR0N EQU	\$FBF7	for FM-7
43	0048	D807		LININP EQU	\$D807	for FM-7 (to \$043D-)
44	0049	00BF		DEVNO EQU	\$BF	for FM-7
45	0050	000D		CR EQU	\$0D	
46	0051	000A		LF EQU	\$0A	
47	0052	0020		SPC EQU	\$20	
48	0053	001B		ESC EQU	\$1B	
49	0054					
50	0055	5E00		RWBUF EQU	STTADR-\$200	
51	0056					
52	0057	02FA		FMDAT EQU	\$02FA	date of FM-7
53	0058	FD9E		PBDAT EQU	\$FD9E	for FM-7
54	0059	FD9F		PBCTL EQU	PBDAT+1	
55	0060	0033		BASADP EQU	\$33	(33,34) pointer for F-
56		BASIC top				
57	0061	0035		BASAEP EQU	\$35	(35,36) pointer for F-
58		BASIC end				
59	0062	2000		TXTADR EQU	\$2000	load adr for TEXT
60	0063	00E5		DELMK EQU	\$E5	delete mark
61	0064					
62	0065	6000		ORG	STTADR	
63	0066					
64	0067	6000	16	0098 START	LBRA	MAIN
65	0068	6003	01	VN	FCB	1
66	0069	6004	00	SCTN0	FCB	0
67	0070	6005	00	SCTN1	FCB	0
68	0071	6006	00	SCTN2	FCB	0
69	0072	6007	00	SCTN3	FCB	0
70	0073	6008	00	RESDAT	FCB	0
71	0074	6009	00	SDHC	FCB	0
72	0075					

73	0076	600A	0000	SCTBPB	FDB	00	sct no of BPB(4bytes)
74	0077	600C	0000		FDB	00	
75	0078	600E	0000	SCTFAT	FDB	00	sct no of
76	FAT#1(4bytes)						
77	0079	6010	0000		FDB	00	
78	0080	6012	0000	SCTDIR	FDB	00	sct no of DIR(4bytes)
79	0081	6014	0000		FDB	00	
80	0082	6016	0000	SCTDSV	FDB	00	save sct no of
81	DIR(4bytes)						
82	0083	6018	0000		FDB	00	
83	0084	601A	0000	SCTCLT	FDB	00	sct no of top
84	CLUSTER(4bytes)						
85	0085	601C	0000		FDB	00	if SD:#2, SDHC:#3
86	0086	601E	00	SCTPCL	FCB	0	sct per cluster
87	0087	601F	FF	DIRENT	RMB	32	directory entry
88	0088	603F	FF	WMDY	RMB	3	mon,day,year
89	0089	6042	FF	WFSIZ	RMB	4	file size(4bytes)
90	0090	6046	FF	FILSIZ	RMB	2	file size
91	0091	6048	0000	CLSTH	FDB	00	cluster High
92	0092	604A	0000	CLSTL	FDB	00	cluster Low
93	0093	604C	00	CNT	FCB	0	
94	0094	604D	00	MCFLG	FCB	0	0:machine code,
95	\$FF,\$FE:binary save 1:else						
96	0095	604E	00	TPGFLG	FCB	0	1:top page 0:else
97	0096	604F	FF	FNAME	RMB	11	file name(8+3)
98	0097	605A	00	DPGNO	FCB	0	page no of empty dir
99	entry						
100	0098	605B	0000	DIROFS	FDB	00	offset of empty dir
101	entry						
102	0099	605D	0000	BASEXP	FDB	00	basic exe adr
103	0100	605F	0000	MCADP	FDB	00	machine code start
104	adr						
105	0101	6061	0000	MCAEP	FDB	00	machine code end adr
106	0102	6063	0000	MCEXP	FDB	00	machine dode exec adr
107	0103	6065	0000	EXSTFL	FDB	00	ofset of exist dir
108	entry						
109	0104	6067	0000	WK1	FDB	00	
110	0105	6069	0000	WK2	FDB	00	
111	0106	606B	0000	WK3	FDB	00	

```

112 0107 606D 0000 WK4 FDB 00
113 0108
114 0109 606F FF INPBUF RMB 8
115 0110 6077 FF CNVBUF RMB 4
116 0111 607B FF FILBUF RMB 32
117 0112
118 0113 609B MAIN EQU *
119 0114 *=====reset pia=====
120 0115 *portB:output
121 0116 609B 86 30 LDA #$30
122 0117 609D B7 FD9F STA PBCTL select ddra, cb2:0
one 0118 60A0 86 FD LDA #$FD pb0,7:out,pb1:in
two 0119 60A2 B7 FD9E STA PBDAT set output mode
three 0120 60A5 86 34 LDA #$34
124 0121 60A7 B7 FD9F STA PBCTL select
125 peripheral,cb2:0
126 0122
127 0123 *for pia exist check
128 0124 60AA 86 55 LDA #$55 test data
129 0125 60AC B7 FD9E STA PBDAT
130 0126 60AF B6 FD9E LDA PBDAT
131 0127 60B2 84 FD ANDA #$FD pb1=0
132 0128 60B4 81 55 CMPA #$55
133 0129 60B6 27 07 BEQ INIT
134 0130 60B8 8E 6F93 LDX #MERR
135 0131 60BB 17 0C68 LBSR PSTRNG
136 0132 60BE 39 RTS Return F-BASIC
137 0133
138 0134 60BF INIT EQU *
139 0135 60BF 17 0091 LBSR SDINIT
140 0136
141 0137 *****
142 0138 * MAIN MENU
143 0139 *****
144 0140 60C2 MENU EQU *
145 0141 60C2 8E 6EAB LDX #MSMENU1
146 0142 60C5 17 0C5E LBSR PSTRNG
147 0143 60C8 BD FBF7 JSR CSRON cursor on
148 0144 60CB BD 6E2D JSR INOUTC

```

149	0145	60CE	84	DF		ANDA	#\$DF	to upper
150	0146							
151	0147	60D0	81	44		CMPA	#'D	
152	0148	60D2	26	05		BNE	MENU1	
153	0149	60D4	17	0328		LBSR	DSPDIR	
154	0150	60D7	20	E9		BRA	MENU	
155	0151	60D9	81	4C	MENU1	CMPA	#'L	
156	0152	60DB	26	05		BNE	MENU2	
157	0153	60DD	17	03CD		LBSR	LDFILE	
158	0154	60E0	20	E0		BRA	MENU	
159	0155	60E2	81	42	MENU2	CMPA	#'B	
160	0156	60E4	26	05		BNE	MENU3	
161	0157	60E6	17	05D8		LBSR	WRBFILE	
162	0158	60E9	20	D7		BRA	MENU	
163	0159	60EB	81	4D	MENU3	CMPA	#'M	
164	0160	60ED	26	05		BNE	MENU4	
165	0161	60EF	17	0641		LBSR	WRMFILE	
166	0162	60F2	20	CE		BRA	MENU	
167	0163	60F4	81	4B	MENU4	CMPA	#'K	
168	0164	60F6	26	05		BNE	MENU5	
169	0165	60F8	17	0571		LBSR	KILFILE	
170	0166	60FB	20	C5		BRA	MENU	
171	0167	60FD	81	53	MENU5	CMPA	#'S	
172	0168	60FF	26	04		BNE	MENU6	
173	0169	6101	20	50		BRA	SDINIT	
174	0170	6103	20	BD		BRA	MENU	
175	0171	6105	81	4F	MENU6	CMPA	#'O	
176	0172	6107	26	04		BNE	MENU7	
177	0173	6109	20	0A		BRA	OPMENU	
178	0174	610B	20	B5		BRA	MENU	
179	0175	610D	81	51	MENU7	CMPA	#'Q	
180	0176	610F	26	B1		BNE	MENU	
181	0177	6111	17	0B5D		LBSR	PCRLF	
182	0178	6114	39			RTS	Return F-BASIC	
183	0179							
184	0180		*					
185	0181	6115		OPMENU	EQU	*		
186	0182	6115	8E	6F01		LDX	#MSMENU2	
187	0183	6118	17	0C0B		LBSR	PSTRNG	

```

188 0184 611B BD 6E2D JSR INOUTC
189 0185 611E 80 30 SUBA #$30
190 0186 6120 26 02 BNE OPMEN6 0?
191 0187 6122 20 9E BRA MENU
192 0188
193 0189 6124 4A OPMEN6 DECA 1?
194 0190 6125 26 05 BNE OPMEN7
195 0191 6127 17 024A LBSR RDSCT
196 0192 612A 20 E9 BRA OPMENU
197 0193 612C 4A OPMEN7 DECA 2?
198 0194 612D 26 05 BNE OPMEN8
199 0195 612F 17 01C1 LBSR DSPSDDAT
200 0196 6132 20 E1 BRA OPMENU
201 0197 6134 4A OPMEN8 DECA 3?
202 0198 6135 26 05 BNE OPMEN9
203 0199 6137 17 024D LBSR DSPMBR
204 0200 613A 20 D9 BRA OPMENU
205 0201 613C 4A OPMEN9 DECA 4?
206 0202 613D 26 05 BNE OPMEN10
207 0203 613F 17 0299 LBSR DSPBPB
208 0204 6142 20 D1 BRA OPMENU
209 0205 6144 4A OPMEN10 DECA 5?
210 0206 6145 26 CE BNE OPMENU
211 0207 6147 17 02A3 LBSR DSPFAT
212 0208 614A 20 C9 BRA OPMENU
213 0209
214 0210 *
215 0211 614C SDERR EQU *
216 0212 614C 8E 6F93 LDX #MERR
217 0213 614F 17 0BD4 LBSR PSTRNG
218 0214 6152 39 RTS Return F-BASIC
219 0215
220 0216 *****
221 0217 * SD Initialize routine
222 0218 *****
223 0219 6153 SDINIT EQU *
224 0220 *----- start -----
225 0221 6153 8E 6F7C LDX #MSINIT
226 0222 6156 17 0BCD LBSR PSTRNG

```

```

227 0223      * cs:1,di:1,sck>74ms
228 0224 6159 86 FF          LDA      #$FF
229 mosi:b7=H,miso:b1=H(for PIA),cs:b0=H
230 0225 615B B7 FD9E          STA      PBDAT
231 0226 615E C6 50          LDB      #80      80times
232 0227 6160 17 0AD3  ISTT    LBSR      SETCKL
233 0228 6163 17 0AD9          LBSR      SETCKH
234 0229 6166 5A          DECB
235 0230 6167 26 F7          BNE      ISTT
236 0231      *
237 0232      *----- cmd 0 -----
238 0233 6169          SCMD0  EQU      *
239 0234 6169 17 0ACA          LBSR      SETCKL
240 0235 616C C6 06          LDB      #6
241 0236 616E 8E 6E47          LDX      #CMD0
242 0237 6171 A6 80 SC01    LDA      ,X+
243 0238 6173 17 0A53          LBSR      SNDDAT
244 0239 6176 5A          DECB
245 0240 6177 26 F8          BNE      SC01
246 0241      *get response
247 0242 6179 5F          CLRB
248 0243 617A          GR00
249 0244 617A 5A          DECB
250 0245 617B 27 CF          BEQ      SDERR
251 0246 617D 17 0A65          LBSR      GETDAT
252 0247 6180 B6 6008          LDA      RESDAT
253 0248 6183 81 01          CMPA      #$01
254 0249 6185 26 F3          BNE      GR00
255 0250      *
256 0251      *----- cmd 8 -----
257 0252 6187          SCMD8  EQU      *
258 0253 6187 17 0AAC          LBSR      SETCKL
259 0254 618A 86 FF          LDA      #$FF      dummy cmd
260 0255 618C 17 0A3A          LBSR      SNDDAT
261 0256 618F C6 06          LDB      #6
262 0257 6191 8E 6E5F          LDX      #CMD8
263 0258 6194 A6 80 SC80    LDA      ,X+
264 0259 6196 17 0A30          LBSR      SNDDAT
265 0260 6199 5A          DECB

```

266	0261	619A	26	F8	BNE	SC80	
267	0262			*get response			
268	0263	619C		GR80			
269	0264	619C	17	0A46	LBSR	GETDAT	
270	0265	619F	B6	6008	LDA	RESDAT	
271	0266						
272	0267	61A2	17	0A40	LBSR	GETDAT	
273	0268	61A5	B6	6008	LDA	RESDAT	
274	0269	61A8	81	01	CMPA	#\$01	
275	0270	61AA	26	F0	BNE	GR80	
276	0271	61AC	C6	04	LDB	#4	
277	0272	61AE	17	0A34	GR81 LBSR	GETDAT	
278	0273	61B1	B6	6008	LDA	RESDAT	
279	0274	61B4	5A		DECB		
280	0275	61B5	26	F7	BNE	GR81	
281	0276			*			
282	0277			*----- cmd 58 -----			
283	0278	61B7		SCMD58 EQU	*		
284	0279	61B7	17	0A7C	LBSR	SETCKL	
285	0280	61BA	86	FF	LDA	#\$FF	dummy cmd
286	0281	61BC	17	0A0A	LBSR	SNDDAT	
287	0282	61BF	C6	06	LDB	#6	
288	0283	61C1	8E	6E65	LDX	#CMD58	
289	0284	61C4	A6	80 SC581	LDA	,X+	
290	0285	61C6	17	0A00	LBSR	SNDDAT	
291	0286	61C9	5A		DECB		
292	0287	61CA	26	F8	BNE	SC581	
293	0288			*get response			
294	0289	61CC		GR580			
295	0290	61CC	17	0A16	LBSR	GETDAT	
296	0291	61CF	B6	6008	LDA	RESDAT	
297	0292	61D2	81	01	CMPA	#\$01	
298	0293	61D4	26	F6	BNE	GR580	
299	0294	61D6	C6	04 GR581	LDB	#4	
300	0295	61D8	17	0A0A GR582	LBSR	GETDAT	
301	0296	61DB	B6	6008	LDA	RESDAT	
302	0297	61DE	5A		DECB		
303	0298	61DF	26	F7	BNE	GR582	
304	0299	61E1		RPT41 EQU	*		

```

305 0300      *----- cmd 55 -----
306 0301 61E1      SCMD55 EQU      *
307 0302 61E1 17 0A52      LBSR      SETCKL
308 0303 61E4 86 FF      LDA      #$FF      dummy cmd
309 0304 61E6 17 09E0      LBSR      SNDDAT
310 0305 61E9 C6 06      LDB      #6
311 0306 61EB 8E 6E6B      LDX      #CMD55
312 0307 61EE A6 80 SC551 LDA      ,X+
313 0308 61F0 17 09D6      LBSR      SNDDAT
314 0309 61F3 5A      DECB
315 0310 61F4 26 F8      BNE      SC551
316 0311      *get response
317 0312      * CLRB
318 0313 61F6      GR550
319 0314 61F6 17 09EC      LBSR      GETDAT
320 0315 61F9 B6 6008      LDA      RESDAT
321 0316 61FC 81 01      CMPA      #$01
322 0317 61FE 26 F6      BNE      GR550
323 0318
324 0319      *----- acmd 41 -----
325 0320 6200      ASCMD41 EQU      *
326 0321 6200 17 0A33      LBSR      SETCKL
327 0322 6203 86 FF      LDA      #$FF      dummy cmd
328 0323 6205 17 09C1      LBSR      SNDDAT
329 0324 6208 C6 06      LDB      #6
330 0325 620A 8E 6E71      LDX      #ACMD41
331 0326 620D A6 80 ASC411 LDA      ,X+
332 0327 620F 17 09B7      LBSR      SNDDAT
333 0328 6212 5A      DECB
334 0329 6213 26 F8      BNE      ASC411
335 0330      *get response
336 0331 6215      GR410
337 0332 6215 17 09CD      LBSR      GETDAT
338 0333 6218 B6 6008      LDA      RESDAT
339 0334 621B 17 09C7      LBSR      GETDAT
340 0335 621E B6 6008      LDA      RESDAT
341 0336 6221 81 00      CMPA      #$00
342 0337 6223 26 BC      BNE      RPT41
343 0338

```

```

344 0339          *----- second cmd 58 -----
345 0340 6225      SCMD258 EQU      *
346 0341 6225 17 0A0E          LBSR      SETCKL
347 0342 6228 86 FF          LDA      #$FF      dummy cmd
348 0343 622A 17 099C          LBSR      SNDDAT
349 0344 622D C6 06          LDB      #6
350 0345 622F 8E 6E65          LDX      #CMD58
351 0346 6232 A6 80 SC2581 LDA      ,X+
352 0347 6234 17 0992          LBSR      SNDDAT
353 0348 6237 5A          DECB
354 0349 6238 26 F8          BNE      SC2581
355 0350          *get response
356 0351 623A      GR2580
357 0352 623A 17 09A8          LBSR      GETDAT
358 0353 623D B6 6008          LDA      RESDAT
359 0354 6240 81 00          CMPA      #$00
360 0355 6242 26 F6          BNE      GR2580
361 0356 6244 17 099E          LBSR      GETDAT
362 0357 6247 B6 6008          LDA      RESDAT
363 0358 624A 48          LSLA
364 0359 624B 48          LSLA
365 0360 624C 24 03          BCC      GR2581
366 0361 624E 7C 6009          INC      SDHC      if sdhc then 1
367 0362 6251 C6 03 GR2581 LDB      #3
368 0363 6253 17 098F GR2582 LBSR      GETDAT
369 0364 6256 B6 6008          LDA      RESDAT
370 0365 6259 5A          DECB
371 0366 625A 26 F7          BNE      GR2582
372 0367
373 0368 625C 8E 6F66          LDX      #MSSD0
374 0369 625F B6 6009          LDA      SDHC
375 0370 6262 27 03          BEQ      SD1
376 0371 6264 8E 6F70          LDX      #MSSD1
377 0372 6267 17 0ABC SD1 LBSR      PSTRNG
378 0373          *
379 0374          *set sct no of BPB
380 0375 626A 17 0138          LBSR      GETMBR
381 0376 626D 8E 5FCA          LDX      #RWBUF+446+8+4
382 0377 6270 108E 600A          LDY      #SCTBPB

```

```

383 0378 6274 C6 04          LDB      #4
384 0379 6276 A6 82  IL0P1  LDA      , -X
385 0380 6278 A7 A0          STA      ,Y+
386 0381 627A 5A          DECB
387 0382 627B 26 F9          BNE      IL0P1
388 0383
389 0384          *set sct no of FAT#1
390 0385 627D 17 0142          LBSR     GETBPB
391 0386 6280 8E 600A          LDX      #SCTBPB
392 0387 6283 108E 600E          LDY      #SCTFAT
393 0388 6287 C6 04          LDB      #4
394 0389 6289 17 08A1          LBSR     TRANS
395 0390 628C 8E 600E          LDX      #SCTFAT
396 0391 628F 108E 5E0E          LDY      #RWBUF+$0E      for SD & SDHC
397 0392 6293 17 0A5F          LBSR     ADD42
398 0393
399 0394          *set sct no of DIR
400 0395 6296 8E 600E          LDX      #SCTFAT
401 0396 6299 108E 6012          LDY      #SCTDIR
402 0397 629D C6 04          LDB      #4
403 0398 629F 17 088B          LBSR     TRANS
404 0399 62A2 8E 5E16          LDX      #RWBUF+$16      for SD
405 0400 62A5 7D 6009          TST      SDHC      0:SD,1:SDHC
406 0401 62A8 27 03          BEQ      SKP1
407 0402 62AA C6 0E          LDB      #$24-$16
408 0403 62AC 3A          ABX          for SDHC
409 0404 62AD 1F 12  SKP1  TFR      X,Y
410 0405 62AF 8E 6012          LDX      #SCTDIR
411 0406 62B2 17 0A40          LBSR     ADD42
412 0407 62B5 17 0A3D          LBSR     ADD42      skip for FAT#2
413 0408
414 0409          *save no of directory
415 0410 62B8 8E 6012          LDX      #SCTDIR
416 0411 62BB 108E 6016          LDY      #SCTDSV
417 0412 62BF C6 04          LDB      #4
418 0413 62C1 17 0869          LBSR     TRANS
419 0414
420 0415          *set sct per cluster
421 0416 62C4 8E 5E0D          LDX      #RWBUF+$0D

```

```

422 0417 62C7 A6 84          LDA      ,X
423 0418 62C9 B7 601E        STA      SCTPCL
424 0419
425 0420          *set sct no of CLUSTER#2
426 0421 62CC 8E 6012        LDX      #SCTDIR
427 0422 62CF 108E 601A      LDY      #SCTCLT
428 0423 62D3 C6 04          LDB      #4
429 0424 62D5 17 0855        LBSR     TRANS
430 0425
431 0426 62D8 108E 6069      LDY      #WK1+2
432 0427 62DC 86 00          LDA      #$00
433 0428 62DE A7 A2          STA      , -Y
434 0429 62E0 86 20          LDA      #$20          for SD
435 0430 62E2 7D 6009        TST      SDHC
436 0431 62E5 27 03          BEQ      SKP2
437 0432 62E7 B6 601E        LDA      SCTPCL          for SDHC
438 0433 62EA A7 A2 SKP2     STA      , -Y
439 0434 62EC 8E 601A        LDX      #SCTCLT
440 0435 62EF 17 0A03        LBSR     ADD42
441 0436 62F2 39            RTS
442 0437
443 0438          *****
444 0439          * GET SD DATA ROUTINE
445 0440          * use CMD9
446 0441          *****
447 0442 62F3          DSPSDDAT EQU      *
448 0443 62F3 17 097B        LBSR     PCRLF
449 0444 62F6 17 093D        LBSR     SETCKL
450 0445 62F9 C6 06          LDB      #6
451 0446 62FB 8E 6E77        LDX      #CMD9
452 0447 62FE A6 80 SC91     LDA      ,X+
453 0448 6300 17 08C6        LBSR     SNDDAT
454 0449 6303 5A            DECB
455 0450 6304 26 F8          BNE      SC91
456 0451          *get response
457 0452 6306          GR90
458 0453 6306 17 08DC        LBSR     GETDAT
459 0454 6309 B6 6008        LDA      RESDAT
460 0455 630C 81 00          CMPA     #$00

```

461	0456	630E	26	F6		BNE	GR90	
462	0457							
463	0458	6310	C6	14		LDB	#16+1+2+1	fe,crcH,crcL,ff
464	0459	6312	17	08D0	GR91	LBSR	GETDAT	
465	0460	6315	B6	6008		LDA	RESDAT	
466	0461	6318	5A			DECB		
467	0462	6319	26	F7		BNE	GR91	
468	0463	631B	17	0953		LBSR	PCRLF	
469	0464							
470	0465						*disp offset of MBR,FAT#1,DIR,CLUSTER#2	
471	0466	631E	8E	6FB3		LDX	#MSOFS	
472	0467	6321	17	0A02		LBSR	PSTRNG	
473	0468	6324	8E	6FC3		LDX	#MSBPB	
474	0469	6327	17	09FC		LBSR	PSTRNG	
475	0470	632A	8E	600A		LDX	#SCTBPB	
476	0471	632D	C6	04		LDB	#4	
477	0472	632F	A6	80	SCL0P1	LDA	,X+	
478	0473	6331	17	0914		LBSR	DSPBIN	
479	0474	6334	5A			DECB		
480	0475	6335	26	F8		BNE	SCL0P1	
481	0476							
482	0477	6337	8E	6FCC		LDX	#MSFAT	
483	0478	633A	17	09E9		LBSR	PSTRNG	
484	0479	633D	8E	600E		LDX	#SCTFAT	
485	0480	6340	C6	04		LDB	#4	
486	0481	6342	A6	80	SCL0P2	LDA	,X+	
487	0482	6344	17	0901		LBSR	DSPBIN	
488	0483	6347	5A			DECB		
489	0484	6348	26	F8		BNE	SCL0P2	
490	0485							
491	0486	634A	8E	6FD5		LDX	#MSDIR	
492	0487	634D	17	09D6		LBSR	PSTRNG	
493	0488	6350	8E	6012		LDX	#SCTDIR	
494	0489	6353	C6	04		LDB	#4	
495	0490	6355	A6	80	SCL0P3	LDA	,X+	
496	0491	6357	17	08EE		LBSR	DSPBIN	
497	0492	635A	5A			DECB		
498	0493	635B	26	F8		BNE	SCL0P3	
499	0494							

```

500 0495 635D 8E 6FDE          LDX      #MSCL2
501 0496 6360 17 09C3          LBSR     PSTRNG
502 0497 6363 8E 601A          LDX      #SCTCLT
503 0498 6366 C6 04            LDB      #4
504 0499 6368 A6 80 SCL0P4 LDA      ,X+
505 0500 636A 17 08DB          LBSR     DSPBIN
506 0501 636D 5A              DECB
507 0502 636E 26 F8            BNE      SCL0P4
508 0503 6370 17 08FE          LBSR     PCRLF
509 0504 6373 39              RTS
510 0505
511 0506      *
512 0507      *****
513 0508      * READ ROUTINE
514 0509      * use CMD17
515 0510      * set sctno to SCTN2,SCTN3
516 0511      * read data set to RWBUF
517 0512      *****
518 0513 6374      RDSCT EQU      *
519 0514 6374 17 08FA          LBSR     PCRLF
520 0515 6377 17 0885          LBSR     INSCT      input sct no
521 0516 637A 17 07F8          LBSR     RD1SCT      read 1 sct
522 0517
523 0518      *disp data
524 0519 637D 17 08F1          LBSR     PCRLF
525 0520 6380 17 0828          LBSR     DSPBUF
526 0521 6383 17 08EB          LBSR     PCRLF
527 0522 6386 39              RTS
528 0523
529 0524      *****
530 0525      * Read MBR & Disp parameter of BPB
531 0526      *****
532 0527 6387      DSPMBR EQU      *
533 0528 6387 17 08E7          LBSR     PCRLF
534 0529 638A 8D 19            BSR      GETMBR
535 0530
536 0531      *disp BPB
537 0532 638C 17 08E2          LBSR     PCRLF
538 0533 638F 8E 5FBE          LDX      #RWBUF+446      1st BPB point

```

```

539 0534 6392 C6 10 GET1 LDB #16
540 0535 6394 A6 80 GET2 LDA ,X+
541 0536 6396 17 08AF LBSR DSPBIN disp data
542 0537 6399 5A DECB
543 0538 639A 26 F8 BNE GET2
544 0539 639C 17 08D2 LBSR PCRLF
545 0540 639F 8C 5FFE CMPX #RWBUF+510 446+16*4
546 0541 63A2 25 EE BLO GET1
547 0542 63A4 39 RTS
548 0543
549 0544 *-----
550 0545 *read MBR
551 0546 63A5 GETMBR EQU *
552 0547 63A5 8E 6004 LDX #SCTN0
553 0548 63A8 C6 04 LDB #4
554 0549 63AA 4F CLRA
555 0550 63AB 17 078A LBSR FILL set sct no is 0
556 0551 63AE 17 0856 LBSR SETSCT
557 0552 63B1 17 07C1 LBSR RD1SCT read 1 sct
558 0553 63B4 39 RTS
559 0554
560 0555 *-----
561 0556 *set sector no to SCTN0
562 0557 *from X,X+1,X+2,X+3 to SCTN0-SCTN3
563 0558 63B5 SETSN0 EQU *
564 0559 63B5 108E 6004 LDY #SCTN0
565 0560 63B9 C6 04 LDB #4
566 0561 63BB 17 076F LBSR TRANS
567 0562 63BE 17 0846 LBSR SETSCT
568 0563 63C1 39 RTS
569 0564
570 0565 *-----
571 0566 *read BPB
572 0567 63C2 GETBPB EQU *
573 0568 63C2 8D E1 BSR GETMBR
574 0569 *set sct no of BPB
575 0570 63C4 8E 5FCA LDX #RWBUF+446+8+4
576 0571 63C7 108E 6004 LDY #SCTN0
577 0572 63CB C6 04 LDB #4

```

```

578 0573 63CD A6 82 GTBLOP LDA , -X
579 0574 63CF A7 A0 STA ,Y+
580 0575 63D1 5A DECB
581 0576 63D2 26 F9 BNE GTBLOP
582 0577 63D4 17 0830 LBSR SETSCT
583 0578 63D7 17 079B LBSR RD1SCT read 1 sct
584 0579 63DA 39 RTS
585 0580
586 0581 *****
587 0582 * Disp BPB(Bios Parameter Block)
588 0583 *****
589 0584 63DB DSPBPB EQU *
590 0585 63DB 8E 600A LDX #SCTBPB
591 0586 63DE 8D D5 BSR SETSNO
592 0587 63E0 17 0792 LBSR RD1SCT read 1 sct
593 0588
594 0589 *disp BPB
595 0590 63E3 17 088B LBSR PCRLF
596 0591 63E6 17 07C2 LBSR DSPBUF
597 0592 63E9 17 0885 LBSR PCRLF
598 0593 63EC 39 RTS
599 0594
600 0595 *****
601 0596 * Disp FAT#1
602 0597 *****
603 0598 63ED DSPFAT EQU *
604 0599 63ED 8E 600E LDX #SCTFAT
605 0600 63F0 8D C3 BSR SETSNO
606 0601 63F2 17 0780 LBSR RD1SCT read 1 sct
607 0602
608 0603 *disp FAT
609 0604 63F5 17 0879 LBSR PCRLF
610 0605 63F8 17 07B0 LBSR DSPBUF
611 0606 63FB 17 0873 LBSR PCRLF
612 0607 63FE 39 RTS
613 0608
614 0609 *****
615 0610 * Disp Directory
616 0611 *****

```

```

617 0612 63FF      DSPDIR EQU      *
618 0613      *return no of DIR
619 0614 63FF 8E 6016      LDX      #SCTDSV
620 0615 6402 108E 6012      LDY      #SCTDIR
621 0616 6406 C6 04      LDB      #4
622 0617 6408 17 0722      LBSR      TRANS
623 0618 640B 8E 6012 DSPFL0 LDX      #SCTDIR
624 0619 640E 8D A5      BSR      SETSNO
625 0620 6410 17 0762      LBSR      RD1SCT      read 1 sct
626 0621
627 0622      *disp FILE name
628 0623 6413 17 085B      LBSR      PCRLF
629 0624 6416 17 0858      LBSR      PCRLF
630 0625 6419 8E 5E00      LDX      #RWBUF
631 0626
632 0627 641C A6 84 DSPFL1 LDA      ,X
633 0628 641E 1027 008A      LBEQ      DSPEND      if top of
634 filename is 0 then end
635 0629 6422 81 E5      CMPA      #$E5      delete mark?
636 0630 6424 27 06      BEQ      DSPCKE
637 0631 6426 A6 0B      LDA      11,X      attr
638 0632      * CMPA #$0F long file name?
639 0633      * BEQ DSPCKE
640 0634      * CMPA #$10 directory?
641 0635      * BEQ DSPCKE
642 0636 6428 81 20      CMPA      #$20
643 0637 642A 27 0A      BEQ      DSPFL2      archive file only
644 0638 642C 30 88 20 DSPCKE LEAX      32,X      next directory
645 entry
646 0639 642F 8C 6000      CMPX      #RWBUF+512      over buf?
647 0640 6432 24 60      BHS      DSPFL5
648 0641 6434 20 E6      BRA      DSPFL1
649 0642
650 0643 6436      DSPFL2
651 0644      *disp filename
652 0645 6436 C6 08      LDB      #8
653 0646 6438 A6 80 DSPFL3 LDA      ,X+
654 0647 643A BD FB BB      JSR      OUTCHA
655 0648 643D 5A      DECB

```

656	0649	643E	26	F8	BNE	DSPFL3	
657	0650			*disp ext			
658	0651	6440	86	2E	LDA	#'.	
659	0652	6442	BD	FBBB	JSR	OUTCHA	
660	0653	6445	C6	03	LDB	#3	
661	0654	6447	A6	80	DSPFL4 LDA	,X+	
662	0655	6449	BD	FBBB	JSR	OUTCHA	
663	0656	644C	5A		DECB		
664	0657	644D	26	F8	BNE	DSPFL4	
665	0658	644F	86	20	LDA	#SPC	
666	0659	6451	BD	FBBB	JSR	OUTCHA	
667	0660	6454	BD	FBBB	JSR	OUTCHA	
668	0661						
669	0662			*get FILE time data			
670	0663	6457	30	07	LEAX	7,X	
671	0664	6459	EC	81	LDD	,X++	
672	0665	645B	FD	6067	STD	WK1	
673	0666						
674	0667			*get CLUSTER(H) data			
675	0668	645E	EC	81	LDD	,X++	
676	0669	6460	1E	89	EXG	A,B	
677	0670	6462	FD	6048	STD	CLSTH	
678	0671	6465	30	04	LEAX	4,X	
679	0672	6467	EC	81	LDD	,X++	
680	0673	6469	1E	89	EXG	A,B	
681	0674	646B	FD	604A	STD	CLSTL	
682	0675						
683	0676			*get FILE size data			
684	0677	646E	108E	6046	LDY	#WFSIZ+4	
685	0678	6472	C6	04	LDB	#4	
686	0679	6474	A6	80	DSPC1 LDA	,X+	
687	0680	6476	A7	A2	STA	, -Y	
688	0681	6478	5A		DECB		
689	0682	6479	26	F9	BNE	DSPC1	
690	0683						
691	0684			*disp FILE size			
692	0685	647B	86	20	LDA	#SPC	
693	0686	647D	BD	FBBB	JSR	OUTCHA	
694	0687	6480	17	0914	LBSR	DSPSIZ	disp FILE size

```

695 0688
696 0689      *disp FILE time
697 0690 6483 FC 6067      LDD      WK1
698 0691 6486 17 08AD      LBSR      DSPYMD      disp MM-DD-YY
699 0692
700 0693      *disp CLUSTER(H,L)
701 0694 6489 17 0960      LBSR      DSPCLS
702 0695 648C 17 07E2      LBSR      PCRLF
703 0696
704 0697 648F 8C 6000      CMPX      #RWBUF+512
705 0698 6492 25 88      BLO      DSPFL1
706 0699 6494 8E 6012 DSPFL5 LDX      #SCTDIR
707 0700 6497 17 06A7      LBSR      INCX4      sct no +1
708 0701 649A 8E 7057      LDX      #MSNXTTP
709 0702 649D 17 0886      LBSR      PSTRNG
710 0703 64A0 BD 6E2D DSPFL6 JSR      INOUTC
711 0704 64A3 81 20      CMPA      #SPC
712 0705 64A5 1027 FF62      LBEQ      DSPFL0
713 0706 64A9 BD 6C71      JSR      PCRLF
714 0707 64AC 39      DSPEND RTS
715 0708
716 0709      *****
717 0710      * Load FILE
718 0711      *****
719 0712 64AD      LDFILE EQU      *
720 0713 64AD 17 0146      LBSR      GETFDT
721 0714 64B0 BE 6065      LDX      EXSTFL      if 0 not exist
722 0715 64B3 1027 00EA      LBEQ      LDNEND
723 0716
724 0717      *get cluster no
725 0718 64B7 EC 88 14      LDD      $14,X
726 0719 64BA 1E 89      EXG      A,B
727 0720 64BC FD 6048      STD      CLSTH      high cluster no
728 0721 64BF EC 88 1A      LDD      $1A,X
729 0722 64C2 1E 89      EXG      A,B
730 0723 64C4 FD 604A      STD      CLSTL      low cluster no
731 0724      *sub start cluster no (SD:2,SDHC:3)
732 0725 64C7 83 0002      SUBD      #2
733 0726 64CA 7D 6009      TST      SDHC      SD:0,SDHC:1

```

734	0727	64CD	27	03		BEQ	LD3	
735	0728	64CF	83	0001		SUBD	#1	D:cluster no (start
736								0)
737	0729		*		(A=0,cluster no is only B)			
738	0730							
739	0731	64D2	34	36	LD3	PSHS	A,B,X,Y	
740	0732	64D4	8E	601A		LDX	#SCTCLT	top of cluster no
741	0733	64D7	108E	6067		LDY	#WK1	
742	0734	64DB	C6	04		LDB	#4	
743	0735	64DD	17	064D		LBSR	TRANS	
744	0736	64E0	35	36		PULS	A,B,X,Y	
745	0737							
746	0738				*calc start cluster no of file			
747	0739				*(B)x40h			
748	0740	64E2	34	30		PSHS	X,Y	
749	0741	64E4	8E	6067		LDX	#WK1	
750	0742	64E7	17	0666		LBSR	ADD40H	
751	0743	64EA	35	30		PULS	X,Y	
752	0744							
753	0745				*get file size			
754	0746	64EC	34	30		PSHS	X,Y	
755	0747	64EE	30	88 20		LEAX	32,X	
756	0748	64F1	108E	6042		LDY	#WFSIZ	
757	0749	64F5	C6	04		LDB	#4	
758	0750	64F7	A6	82	LD41	LDA	, -X	
759	0751	64F9	A7	A0		STA	,Y+	
760	0752	64FB	5A			DECB		
761	0753	64FC	26	F9		BNE	LD41	
762	0754	64FE	35	30		PULS	X,Y	
763	0755							
764	0756	6500	EC	88 1D		LDD	\$1D,X	upper of file size
765	0757	6503	1E	89		EXG	A,B	
766	0758	6505	44			LSRA	wfsiz/\$200+1	
767	0759	6506	56			RORB		
768	0760	6507	C3	0001		ADDD	#1	
769	0761							
770	0762	650A	C1	40		CMPB	#\$40	40h page?
771	0763	650C	23	02		BLS	LD5	
772	0764	650E	C6	40		LDB	#\$40	

773	0765	6510	F7	604C	LD5	STB	CNT	sct counter
774	0766	6513	F7	604E		STB	TPGFLG	set top page
775	0767							
776	0768							*set start sct no of file
777	0769	6516	34	12		PSHS	A,X	
778	0770	6518	8E	7016		LDX	#MSLDNG	
779	0771	651B	17	0808		LBSR	PSTRNG	
780	0772	651E	35	12		PULS	A,X	
781	0773	6520				LDL0P		
782	0774	6520	86	2E		LDA	#'.	
783	0775	6522	BD	FBBB		JSR	OUTCHA	
784	0776	6525	8E	6067		LDX	#WK1	
785	0777	6528	108E	6004		LDY	#SCTN0	
786	0778	652C	C6	04		LDB	#4	
787	0779	652E	17	05FC		LBSR	TRANS	
788	0780	6531	17	06D3		LBSR	SETSCT	
789	0781	6534	17	063E		LBSR	RD1SCT	read 1 file data
790	0782							
791	0783	6537	7D	604E		TST	TPGFLG	top sector page?
792	0784	653A	27	3B		BEQ	LD9	
793	0785							
794	0786							*at top sector, ini set
795	0787							*offset 00 data is
796	0788							* \$00 :machine code
797	0789							* \$FF,\$FE:BASIC PROGRAM (binary save)
798	0790							* else :general text
799	0791	653C	10BE	6044		LDY	WFSIZ+2	
800	0792	6540	10BF	6046		STY	FILSIZ	set file size
801	0793							
802	0794							* LDX FILSIZ
803	0795							* LEAX 1,X
804	0796							* STX BASAEP
805	0797							
806	0798	6544	8E	5E00		LDX	#RWBUF	
807	0799	6547	A6	84		LDA	,X	
808	0800	6549	B7	604D		STA	MCFLG	
809	0801	654C	27	04		BEQ	LD8	
810	0802	654E	2A	24		BPL	LD81	
811	0803	6550	2B	1E		BMI	LD80	

812	0804					*at machine code, read header		
813	0805	6552	10AE	01	LD8	LDY	1,X	get file size
814	0806	6555	10BF	6046		STY	FILSIZ	
815	0807	6559	EE	03		LDU	3,X	get store addr
816	0808							*offset?
817	0809	655B	8E	70A4		LDX	#MSMCOF	
818	0810	655E	17	07C5		LBSR	PSTRNG	
819	0811	6561	BD	6E2D		JSR	INOUTC	
820	0812	6564	84	DF		ANDA	#\$DF	
821	0813	6566	81	59		CMPA	#'Y	
822	0814	6568	26	0D		BNE	LD9	
823	0815	656A	33C9	E000		LEAU	-\$2000,U	
824	0816	656E	20	07		BRA	LD9	
825	0817							
826	0818	6570	DE	33	LD80	LDU	BASADP	basic file
827	0819	6572	20	03		BRA	LD9	
828	0820	6574			LD81			
829	0821	6574	CE	2000		LDU	#TXTADR	(else) not basic
830								file
831	0822	6577			LD9			
832	0823	6577	8D	30		BSR	MEMSET	expand memory from
833								rwbuf
834	0824							
835	0825	6579	7F	604E		CLR	TPGFLG	end of top page
836	0826	657C	8E	6067		LDX	#WK1	
837	0827	657F	17	05BF		LBSR	INCX4	sct no +1
838	0828	6582	7A	604C		DEC	CNT	
839	0829	6585	26	99		BNE	LDL0P	
840	0830							
841	0831	6587			LDEND			
842	0832	6587	7D	604D		TST	MCFLG	
843	0833	658A	27	0B		BEQ	LDE0	
844	0834	658C	2A	09		BPL	LDE0	
845	0835	658E	FC	6044		LDD	WFSIZ+2	
846	0836	6591	DB	34		ADDB	BASADP+1	
847	0837	6593	99	33		ADCA	BASADP	
848	0838	6595	DD	35		STD	BASAEP	end adrs
849	0839	6597	8E	702E	LDE0	LDX	#MSLDED	
850	0840	659A	17	0789		LBSR	PSTRNG	

851	0841	659D	17	06D1	LDE1	LBSR	PCRLF	
852	0842	65A0	39			RTS		
853	0843							
854	0844	65A1	8E	7049	LDNEND	LDX	#MSNOFL	
855	0845	65A4	17	077F		LBSR	PSTRNG	
856	0846	65A7	20	F4		BRA	LDE1	
857	0847							
858	0848				*expand memory from rwbuf			
859	0849				*X:rwbuf, U:memory adr, Y:filesize			
860	0850	65A9			MEMSET	EQU	*	
861	0851	65A9	34	30		PSHS	X,Y	
862	0852	65AB	8E	5E00		LDX	#RWBUF	
863	0853	65AE	10BE	6046		LDY	FILSIZ	
864	0854	65B2	7D	604E		TST	TPGFLG	top page?
865	0855	65B5	27	0F		BEQ	MSLOP	
866	0856	65B7	7D	604D		TST	MCFLG	0,\$FF(\$FE),etc
867	0857	65BA	27	04		BEQ	MS0	
868	0858	65BC	2A	08		BPL	MSLOP	etc
869	0859	65BE	2B	04		BMI	MS1	basic file
870	0860				* machine code			
871	0861	65C0	30	05	MS0	LEAX	5,X	skip header
872	0862				* LEAY -5,Y			
873	0863	65C2	20	02		BRA	MSLOP	
874	0864	65C4	30	03	MS1	LEAX	3,X	skip 3 FFs
875	0865				* LEAY -3,Y			
876	0866							
877	0867	65C6	A6	80	MSLOP	LDA	,X+	
878	0868	65C8	A7	C0		STA	,U+	
879	0869	65CA	31	3F		LEAY	-1,Y	
880	0870	65CC	27	05		BEQ	MS3	
881	0871	65CE	8C	6000		CMPL	#RWBUF+512	
882	0872	65D1	25	F3		BLO	MSLOP	
883	0873	65D3	10BF	6046	MS3	STY	FILSIZ	
884	0874	65D7	35	B0		PULS	X,Y,PC	
885	0875							
886	0876				*-----			
887	0877				*X:top of FILE NAME			
888	0878				*return C=1:match,C=0:not match			
889	0879	65D9			CMPL	EQU	*	

890	0880	65D9	34	10		PSHS	X	
891	0881	65DB	108E	604F		LDY	#FNAME	
892	0882	65DF	C6	08		LDB	#8	
893	0883	65E1	A6	80	CMPLOP	LDA	,X+	
894	0884	65E3	81	E5		CMPA	#\$E5	delete mark?
895	0885	65E5	27	0B		BEQ	NOTM	
896	0886	65E7	A1	A0		CMPA	,Y+	
897	0887	65E9	26	07		BNE	NOTM	
898	0888	65EB	5A			DECB		
899	0889	65EC	26	F3		BNE	CMPLOP	
900	0890	65EE	1A	01	MATCH	ORCC	#\$01	C=1
901	0891	65F0	20	02		BRA	CMPEND	
902	0892	65F2	1C	FE	NOTM	ANDCC	#\$FE	C=0
903	0893	65F4	35	90	CMPEND	PULS	X,PC	
904	0894							
905	0895							
906	0896							
907	0897							
908	0898							
909	0899							
910	0900	65F6			GETFDT EQU	*		
911	0901	65F6	34	02		PSHS	A	
912	0902	65F8	8E	0000		LDX	#0	
913	0903	65FB	BF	6065		STX	EXSTFL	
914	0904							
915	0905	65FE	8E	6016		LDX	#SCTDSV	
916	0906	6601	108E	6012		LDY	#SCTDIR	
917	0907	6605	C6	04		LDB	#4	
918	0908	6607	17	0523		LBSR	TRANS	
919	0909							
920	0910	660A	35	02		PULS	A	
921	0911	660C	81	4D		CMPA	#'M	
922	0912	660E	26	05		BNE	GTF1	
923	0913	6610	8E	6FFE		LDX	#MSMSV	
924	0914	6613	20	03		BRA	GTF2	
925	0915	6615	8E	6FEE	GTF1	LDX	#MSINFN	
926	0916	6618	17	070B	GTF2	LBSR	PSTRNG	
927	0917	661B	17	0162		LBSR	INPFLDAT	input file name
928	0918							

```

929      0919          *set filename
930      0920 661E 8E 607B          LDX      #FILBUF
931      0921 6621 17 00E9          LBSR      SETFNM
932      0922
933      0923 6624 8E 6012 LDFL0    LDX      #SCTDIR
934      0924 6627 17 FD8B          LBSR      SETSNO
935      0925 662A 17 0548          LBSR      RD1SCT      read 1 sct
936      0926
937      0927 662D 8E 5E00          LDX      #RWBUF
938      0928 6630 A6 84 LDFL1    LDA      ,X
939      0929 6632 27 37          BEQ      LDFLE      if top of filename is
940      0 then end
941      0930 6634 81 E5          CMPA      #$E5      delete mark?
942      0931 6636 26 10          BNE      LD1
943      0932 6638 A6 0B          LDA      11,X      attr
944      0933 663A 81 20          CMPA      #$20      archive file only
945      0934 663C 27 0A          BEQ      LD1
946      0935 663E 30 88 20          LEAX      32,X      next directory
947      entry
948      0936 6641 8C 6000          CMPX      #RWBUF+512  over buf?
949      0937 6644 24 0E          BHS      LDFL5
950      0938 6646 20 E8          BRA      LDFL1
951      0939
952      0940          *check file name
953      0941 6648          LD1
954      0942 6648 8D 8F          BSR      CMPFN      C=1:match
955      0943 664A 25 10          BCS      LD2
956      0944 664C 30 88 20          LEAX      32,X
957      0945 664F 8C 6000          CMPX      #RWBUF+512
958      0946 6652 25 DC          BLO      LDFL1
959      0947
960      0948 6654          LDFL5
961      0949 6654 8E 6012          LDX      #SCTDIR
962      0950 6657 17 04E7          LBSR      INCX4      sct no +1
963      0951 665A 20 C8          BRA      LDFL0
964      0952
965      0953          *hit file, get dirctory entry
966      0954 665C BF 6065 LD2      STX      EXSTFL      adr of dir entry
967      0955 665F 108E 601F          LDY      #DIRENT

```

```

968 0956 6663 C6 20          LDB      #32
969 0957 6665 17 04C5        LBSR     TRANS
970 0958 6668 8E 601F        LDX      #DIRENT
971 0959 666B 39      LDFLE    RTS
972 0960
973 0961      *
974 0962      *****
975 0963      * KILL FILE
976 0964      *****
977 0965 666C      KILFILE EQU      *
978 0966 666C 8D 88          BSR      GETFDT
979 0967 666E BE 6065        LDX      EXSTFL      if 0 not exist
980 0968 6671 27 5B          BEQ      NOFEND
981 0969
982 0970 6673 8E 7074        LDX      #MSKILL
983 0971 6676 17 06AD        LBSR     PSTRNG
984 0972 6679 BD 6E2D        JSR      INOUTC
985 0973 667C 84 DF          ANDA     #$DF
986 0974 667E 81 59          CMPA     #'Y
987 0975 6680 26 3E          BNE      KILEND
988 0976 6682 8D 01          BSR      KILSUB
989 0977 6684 39            RTS
990 0978
991 0979      *-----
992 0980 6685      KILSUB EQU      *
993 0981      *write dir
994 0982 6685 BE 6065        LDX      EXSTFL      top adr of dir
995 entry
996 0983 6688 EC 88 1A        LDD      $1A,X      get clstlD
997 0984 668B 1E 89          EXG      A,B          B:cluster
998 0985 668D FD 604A        STD      CLSTL
999 0986 6690 86 E5          LDA      #$E5      delete mark
1000 0987 6692 A7 84          STA      ,X
1001 0988 6694 17 03D0        LBSR     WRTSD1
1002 0989
1003 0990      *get fat sct & write
1004 0991 6697 8E 600E        LDX      #SCTFAT
1005 0992 669A 17 FD18        LBSR     SETSNO
1006 0993 669D 17 04D5        LBSR     RD1SCT      read 1 sct

```

```

1007 0994
1008 0995 66A0 FC 604A LDD CLSTL B:cluster
1009 0996 66A3 58 LSLB
1010 0997 66A4 7D 6009 TST SDHC sd:0,sdhc:1
1011 0998 66A7 27 01 BEQ KL1
1012 0999 66A9 58 LSLB
1013 1000 66AA 8E 5E00 KL1 LDX #RWBUF
1014 1001 66AD 30 85 LEAX B,X
1015 1002 66AF C6 02 LDB #2
1016 1003 66B1 7D 6009 TST SDHC sd:0,sdhc:1
1017 1004 66B4 27 02 BEQ KL2
1018 1005 66B6 C6 04 LDB #4
1019 1006 66B8 6F 80 KL2 CLR ,X+
1020 1007 66BA 5A DECB
1021 1008 66BB 26 FB BNE KL2
1022 1009 66BD 17 03A7 LBSR WRTSD1
1023 1010 66C0 KILEND
1024 1011 66C0 39 RTS
1025 1012
1026 1013 *
1027 1014 *****
1028 1015 * Write BASIC FILE
1029 1016 *****
1030 1017 66C1 WRBFILE EQU *
1031 1018 *basic file exist check
1032 1019 66C1 DC 35 LDD BASAEP
1033 1020 66C3 93 33 SUBD BASADP
1034 1021 66C5 1083 0002 CMPD #2
1035 1022 66C9 22 0A BHI WRB0 exist
1036 1023 66CB 17 05A3 LBSR PCRLF
1037 1024 66CE 8E 7049 NOFEND LDX #MSNOFL
1038 1025 66D1 17 0652 LBSR PSTRNG
1039 1026 66D4 39 RTS
1040 1027
1041 1028 *save BASIC program file(binary saved)
1042 1029 66D5 86 FF WRB0 LDA #$FF
1043 1030 66D7 B7 604D STA MCFLG set basic file
1044 1031 *input file name
1045 1032 66DA 17 FF19 LBSR GETFDT

```

1046	1033	66DD	BE	6065	LDX	EXSTFL	if 0 not exist
1047	1034	66E0	27	11	BEQ	WRB1	
1048	1035						
1049	1036						*confirm over write
1050	1037	66E2	8E	7087	LDX	#MSEXST	
1051	1038	66E5	17	063E	LBSR	PSTRNG	
1052	1039	66E8	BD	6E2D	JSR	INOUTC	
1053	1040	66EB	84	DF	ANDA	#\$DF	
1054	1041	66ED	81	59	CMPA	#'Y	
1055	1042	66EF	26	1B	BNE	WRBEND	
1056	1043						*delete exist file
1057	1044	66F1	8D	92	BSR	KILSUB	
1058	1045						
1059	1046	66F3				WRB1	
1060	1047						*set filename
1061	1048	66F3	8E	607B	LDX	#FILBUF	
1062	1049	66F6	8D	15	BSR	SETFNM	
1063	1050						
1064	1051						*adrs
1065	1052						* BASADP start adr
1066	1053						* BASAEP end adr
1067	1054						* BASEXP exec adr
1068	1055						
1069	1056						*main of write routine
1070	1057						*clear dir entry
1071	1058	66F8	8D	2B	BSR	CLRDIR	
1072	1059						
1073	1060	66FA	17	0103	LBSR	SRCFAT	search empty
1074							cluster on FAT
1075	1061	66FD	17	0158	LBSR	SRCDIR	search empty DIR
1076							entry
1077	1062	6700	17	0194	LBSR	MKFLDT	make file data
1078	1063	6703	17	0219	LBSR	WRTFIL	write file
1079	1064	6706	17	03BB	LBSR	WRTDIR	write DIR entry
1080	1065	6709	17	03E5	LBSR	WRTFAT	set FAT
1081	1066	670C				WRBEND	
1082	1067	670C	39		RTS		
1083	1068						
1084	1069						*-----

```

1085 1070      *set filename
1086 1071 670D      SETFNM EQU      *
1087 1072      *clear buf
1088 1073 670D 108E 605A      LDY      #FNAME+11
1089 1074 6711 C6 0B      LDB      #11
1090 1075 6713 86 20      LDA      #$20
1091 1076 6715 A7 A2 WRFLP2 STA      ,-Y
1092 1077 6717 5A      DECB
1093 1078 6718 26 FB      BNE      WRFLP2
1094 1079
1095 1080      *store filename
1096 1081 671A A6 80 WRFL1 LDA      ,X+
1097 1082 671C 81 30      CMPA     #$30
1098 1083 671E 25 04      BLO      WRFL2      until delimiter
1099 1084 6720 A7 A0      STA      ,Y+
1100 1085 6722 20 F6      BRA      WRFL1
1101 1086 6724      WRFL2
1102 1087 6724 39      RTS
1103 1088
1104 1089      *-----
1105 1090      *clear dir entry
1106 1091 6725      CLRDIR EQU      *
1107 1092 6725 8E 601F      LDX      #DIRENT
1108 1093 6728 C6 20      LDB      #32
1109 1094 672A 4F      CLRA
1110 1095 672B 17 040A      LBSR      FILL
1111 1096 672E 86 20      LDA      #$20      set attr
1112 1097 6730 A7 0B      STA      $0B,X
1113 1098 6732 39      RTS
1114 1099
1115 1100      *
1116 1101      *****
1117 1102      * Write MACHINE FILE
1118 1103      *****
1119 1104 6733      WRMFILE EQU      *
1120 1105      *save machine code
1121 1106      *filename,topadr,endadr,exeadr ...machine file
1122 1107 6733 17 FEC0      LBSR      GETFDT
1123 1108 6736 BE 6065      LDX      EXSTFL      if 0 not exist

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1124	1109	6739	27	12	BEQ	WRM1	
1125	1110						
1126	1111						*confirm over write
1127	1112	673B	8E	7087	LDX	#MSEXST	
1128	1113	673E	17	05E5	LBSR	PSTRNG	
1129	1114	6741	BD	6E2D	JSR	INOUTC	
1130	1115	6744	84	DF	ANDA	#\$DF	
1131	1116	6746	81	59	CMPA	#'Y	
1132	1117	6748	26	35	BNE	WRMEND	
1133	1118						*delete exist file
1134	1119	674A	17	FF38	LBSR	KILSUB	
1135	1120						
1136	1121	674D				WRM1	
1137	1122	674D	7F	604D	CLR	MCFLG	set machine code
1138	1123						
1139	1124						*set filename
1140	1125	6750	8E	607B	LDX	#FILBUF	
1141	1126	6753	8D	B8	BSR	SETFNM	
1142	1127						
1143	1128						*set top adr
1144	1129	6755	8D	46	BSR	ASC2BIN	
1145	1130	6757	10BF	605F	STY	MCADP	start adr
1146	1131	675B	8D	30	BSR	SRCDLM	
1147	1132	675D	8D	3E	BSR	ASC2BIN	
1148	1133	675F	10BF	6061	STY	MCAEP	end adr
1149	1134	6763	8D	28	BSR	SRCDLM	
1150	1135	6765	8D	36	BSR	ASC2BIN	
1151	1136	6767	10BF	6063	STY	MCEXP	exec adr
1152	1137						
1153	1138						*main of write routine
1154	1139						*clear dir entry
1155	1140	676B	8D	B8	BSR	CLRDIR	
1156	1141						
1157	1142	676D	17	0090	LBSR	SRCFAT	search empty
1158							cluster on FAT
1159	1143	6770	17	00E5	LBSR	SRCDIR	search empty DIR
1160							entry
1161	1144	6773	17	0121	LBSR	MKFLDT	make file data
1162	1145	6776	17	01A6	LBSR	WRTFIL	write file

1163	1146	6779	17	0348	LBSR	WRTDIR	write DIR entry
1164	1147	677C	17	0372	LBSR	WRTFAT	set FAT
1165	1148	677F		WRMEND			
1166	1149	677F	39		RTS		
1167	1150						
1168	1151			*-----			
1169	1152			*input filename & parameter to filbuf			
1170	1153	6780		INPFLDAT EQU *			
1171	1154	6780	34	24	PSHS	B,Y	
1172	1155	6782	108E	607B	LDY	#FILBUF	
1173	1156	6786	C6	1E	LDB	#30	
1174	1157	6788	17	06A9	LBSR	LININ	
1175	1158	678B	35	A4	PULS	B,Y,PC	
1176	1159						
1177	1160			*-----			
1178	1161			*search delimiter			
1179	1162			*return X:point next char of delimiter			
1180	1163	678D		SRCDLM EQU *			
1181	1164	678D	30	1F	LEAX	-1,X	
1182	1165	678F	A6	80	SRDLM1 LDA	,X+	
1183	1166	6791	81	30	CMPA	#\$30	
1184	1167	6793	24	FA	BHS	SRDLM1	
1185	1168	6795	39		RTS		
1186	1169						
1187	1170			*-----			
1188	1171			*skip delimiter			
1189	1172			*return X:point next char of delimiter			
1190	1173	6796		SKPDLM EQU *			
1191	1174	6796	A6	80	LDA	,X+	
1192	1175	6798	81	30	CMPA	#\$30	
1193	1176	679A	25	FA	BLO	SKPDLM	
1194	1177	679C	39		RTS		
1195	1178						
1196	1179			*-----			
1197	1180			*X - X+3, return Y			
1198	1181	679D		ASC2BIN EQU *			
1199	1182			*clr buf			
1200	1183	679D	34	10	PSHS	X	
1201	1184	679F	8E	6077	LDX	#CNVBUF	

1202	1185	67A2	86	30	LDA	#\$30	
1203	1186	67A4	C6	04	LDB	#4	
1204	1187	67A6	17	038F	LBSR	FILL	
1205	1188	67A9	35	10	PULS	X	
1206	1189						
1207	1190				*search next delimiter		
1208	1191				*return B:char count		
1209	1192	67AB	34	10	PSHS	X	
1210	1193	67AD	5F		CLRB		
1211	1194	67AE	5C		SRL0P1	INCB	
1212	1195	67AF	A6	80	LDA	,X+	
1213	1196	67B1	81	30	CMPA	#\$30	count over \$30
1214	1197	67B3	24	F9	BHS	SRL0P1	
1215	1198	67B5	5A		DECB		
1216	1199	67B6	35	10	PULS	X	
1217	1200						
1218	1201				*trans		
1219	1202	67B8	108E	6077	LDY	#CNVBUF	
1220	1203	67BC	34	04	PSHS	B	
1221	1204	67BE	C6	04	LDB	#4	
1222	1205	67C0	E0	E4	SUBB	,S	
1223	1206	67C2	31	A5	LEAY	B,Y	
1224	1207	67C4	35	04	PULS	B	
1225	1208	67C6	A6	80	SRL0P2	LDA	,X+
1226	1209	67C8	A7	A0	STA	,Y+	
1227	1210	67CA	5A		DECB		
1228	1211	67CB	26	F9	BNE	SRL0P2	
1229	1212						
1230	1213				*convert to bin		
1231	1214	67CD	34	10	PSHS	X	
1232	1215	67CF	8E	6077	LDX	#CNVBUF	
1233	1216	67D2	EC	81	LDD	,X++	
1234	1217	67D4	8D	0F	BSR	CNVSUB	
1235	1218	67D6	34	02	PSHS	A	
1236	1219	67D8	EC	81	LDD	,X++	
1237	1220	67DA	8D	09	BSR	CNVSUB	
1238	1221	67DC	1F	89	TFR	A,B	
1239	1222	67DE	35	02	PULS	A	
1240	1223	67E0	1F	02	TFR	D,Y	

1241	1224	67E2	35	10		PULS	X	
1242	1225	67E4	39			RTS		
1243	1226							
1244	1227				*cnv sub			
1245	1228				*return A			
1246	1229	67E5			CNVSUB	EQU	*	
1247	1230	67E5	80	30		SUBA	#\$30	
1248	1231	67E7	81	11		CMPA	#\$11	
1249	1232	67E9	25	02		BLO	CNV1	
1250	1233	67EB	80	07		SUBA	#7	
1251	1234	67ED	48		CNV1	LSLA		
1252	1235	67EE	48			LSLA		
1253	1236	67EF	48			LSLA		
1254	1237	67F0	48			LSLA		
1255	1238	67F1	C0	30		SUBB	#\$30	
1256	1239	67F3	C1	11		CMPB	#\$11	
1257	1240	67F5	25	02		BLO	CNV2	
1258	1241	67F7	C0	07		SUBB	#7	
1259	1242	67F9	C4	0F	CNV2	ANDB	#\$0F	
1260	1243	67FB	34	04		PSHS	B	
1261	1244	67FD	AB	E0		ADDA	,S+	
1262	1245	67FF	39			RTS		
1263	1246							
1264	1247				*-----			
1265	1248				* search empty area in FAT			
1266	1249	6800			SRCFAT	EQU	*	
1267	1250	6800	8E	600E		LDX	#SCTFAT	
1268	1251	6803	17	FBAF		LBSR	SETSN0	
1269	1252	6806	17	036C		LBSR	RD1SCT	read 1 sct
1270	1253							
1271	1254	6809	8E	5E00		LDX	#RWBUF	
1272	1255				*search \$00			
1273	1256	680C			SRC0			
1274	1257	680C	C6	02		LDB	#2	SD
1275	1258	680E	7D	6009		TST	SDHC	0:SD,1:SDHC
1276	1259	6811	27	02		BEQ	SRCL0P	
1277	1260	6813	C6	04		LDB	#4	SDHC
1278	1261	6815	8D	32	SRCL0P	BSR	SRC00	search 1st 00 ...1st
1279	00							

1280	1262	6817	24	2F	BCC	SFATED	not exist empty area
1281	1263	6819	A6	80	LDA	,X+	next is 00? ..2nd 00
1282	1264	681B	26	EF	BNE	SRC0	restart
1283	1265	681D	7D	6009	TST	SDHC	0:SD,1:SDHC
1284	1266	6820	27	0A	BEQ	SETX	
1285	1267	6822	A6	80	LDA	,X+	next is 00? ..3rd 00
1286	1268	6824	26	E6	BNE	SRC0	restart
1287	1269	6826	A6	80	LDA	,X+	next is 00? ..4th 00
1288	1270	6828	26	E2	BNE	SRC0	restart
1289	1271	682A	30	1E	LEAX	-2,X	
1290	1272	682C			SETX		
1291	1273	682C	30	1E	LEAX	-2,X	pos in RWBUF
1292	1274	682E	3089	A200	LEAX	-RWBUF,X	relative pos
1293	1275	6832	1F	10	TFR	X,D	
1294	1276	6834	44		LSRA		
1295	1277	6835	56		RORB	1/2	
1296	1278	6836	7D	6009	TST	SDHC	0:SD,1:SDHC
1297	1279	6839	27	02	BEQ	SETCL	
1298	1280	683B	44		LSRA		
1299	1281	683C	56		RORB	1/2	
1300	1282	683D	FD	604A	SETCL	STD	CLSTL
1301	1283	6840	1E	89	EXG	A,B	
1302	1284	6842	8E	601F	LDX	#DIRENT	
1303	1285	6845	ED	88 1A	STD	\$1A,X	set cluster L to
1304							dirent
1305	1286	6848	39		SFATED	RTS	
1306	1287						
1307	1288				*search	00 in FAT	
1308	1289	6849	8C	6000	SRC00	CMPX	#RWBUF+512
1309	1290	684C	24	07	BHS	NOEXST	
1310	1291	684E	A6	80	LDA	,X+	
1311	1292	6850	26	F7	BNE	SRC00	
1312	1293	6852	1A	01	EXIST	ORCC	#\$01 set C
1313	1294	6854	39		RTS		
1314	1295	6855	1C	FE	NOEXST	ANDCC	#\$FE reset C
1315	1296	6857	39		RTS		
1316	1297						
1317	1298				*-----		
1318	1299				* search empty DIR entry		

```

1319 1300      * return:DPGNO,DIROFS
1320 1301 6858      SRCDIR EQU      *
1321 1302 6858 7F 605A      CLR      DPGNO
1322 1303      *return no of DIR
1323 1304 685B 8E 6016      LDX      #SCTDSV
1324 1305 685E 108E 6012      LDY      #SCTDIR
1325 1306 6862 C6 04      LDB      #4
1326 1307 6864 17 02C6      LBSR      TRANS
1327 1308 6867 8E 6012 SRCFL0 LDX      #SCTDIR
1328 1309 686A 17 FB48 SRCFL1 LBSR      SETSNO
1329 1310 686D 17 0305      LBSR      RD1SCT      read 1 sct
1330 1311
1331 1312 6870 8E 5E00      LDX      #RWBUF
1332 1313 6873 8C 6000 DLOP1 CMPX      #RWBUF+512
1333 1314 6876 24 0D      BHS      SRCFL5
1334 1315 6878 A6 80      LDA      ,X+
1335 1316 687A 27 15      BEQ      DIRGET
1336 1317 687C 81 E5      CMPA      #$E5      delete mark?
1337 1318 687E 27 11      BEQ      DIRGET
1338 1319 6880 30 88 1F      LEAX      32-1,X
1339 1320 6883 20 EE      BRA      DLOP1
1340 1321
1341 1322 6885 7C 605A SRCFL5 INC      DPGNO
1342 1323 6888 8E 6012      LDX      #SCTDIR
1343 1324 688B 17 02B3      LBSR      INCX4      sct no +1
1344 1325 688E 20 DA      BRA      SRCFL1
1345 1326 6890      DSPED
1346 1327 6890 39      RTS
1347 1328
1348 1329 6891 30 1F DIRGET LEAX      -1,X
1349 1330 6893 BF 605B      STX      DIROFS
1350 1331 6896 39      RTS
1351 1332
1352 1333      *-----
1353 1334 6897      MKFLDT EQU      *
1354 1335      *cluster no & attr already set
1355 1336      *file name input&set
1356 1337 6897 8E 604F      LDX      #FNAME
1357 1338 689A 108E 601F      LDY      #DIRENT

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1358	1339	689E	C6	0B		LDB	#11	
1359	1340	68A0	17	028A		LBSR	TRANS	set file name
1360	1341							
1361	1342							*cal file size
1362	1343	68A3	7D	604D		TST	MCFLG	00,FF(FE),else
1363	1344	68A6	2B	08		BMI	MKF1	
1364	1345	68A8	FC	6061		LDD	MCAEP	
1365	1346	68AB	B3	605F		SUBD	MCADP	
1366	1347	68AE	20	04		BRA	MKF2	
1367	1348	68B0						MKF1
1368	1349	68B0	DC	35		LDD	BASAEP	
1369	1350	68B2	93	33		SUBD	BASADP	
1370	1351	68B4	1E	89	MKF2	EXG	A,B	
1371	1352	68B6	8E	601F		LDX	#DIRENT	
1372	1353	68B9	ED	88	1C	STD	\$1C,X	
1373	1354							
1374	1355							*set date (for FM-7)
1375	1356	68BC	8D	12		BSR	STDAT	
1376	1357	68BE	FC	603F		LDD	WMDY	
1377	1358	68C1	1E	89		EXG	A,B	
1378	1359							
1379	1360							*year,month,day
1380	1361	68C3	8E	601F		LDX	#DIRENT	
1381	1362	68C6	ED	88	18	STD	\$18,X	
1382	1363	68C9	ED	88	10	STD	\$10,X	
1383	1364	68CC	ED	88	12	STD	\$12,X	
1384	1365	68CF	39			RTS		
1385	1366							
1386	1367							*-----
1387	1368							*FM day to SD
1388	1369							*retuen WMDY,WMDY+1
1389	1370	68D0			STDAT EQU	*		
1390	1371	68D0	8E	02FA		LDX	#FMDAT	date of FM-7
1391	1372	68D3	108E	603F		LDY	#WMDY	
1392	1373	68D7	C6	03		LDB	#3	
1393	1374	68D9	34	04	ABLOP	PSHS	B	
1394	1375	68DB	EC	81		LDD	,X++	
1395	1376	68DD	8D	2E		BSR	APLUSB	
1396	1377	68DF	A7	A0		STA	,Y+	

1397	1378	68E1	35	04	PULS	B	
1398	1379	68E3	26	F4	BNE	ABL0P	
1399	1380		*year				
1400	1381	68E5	B6	603F	LDA	WMDY	
1401	1382	68E8	8B	14	ADDA	#20	---2000-1980
1402	1383	68EA	48		LSLA		
1403	1384	68EB	B7	603F	STA	WMDY	
1404	1385		*month				
1405	1386	68EE	B6	6040	LDA	WMDY+1	
1406	1387	68F1	48		LSLA		
1407	1388	68F2	48		LSLA		
1408	1389	68F3	48		LSLA		
1409	1390	68F4	48		LSLA		
1410	1391	68F5	48		ASLA		
1411	1392	68F6	B7	6040	STA	WMDY+1	
1412	1393	68F9	F6	603F	LDB	WMDY	
1413	1394	68FC	C9	00	ADCB	#0	plus C
1414	1395	68FE	F7	603F	STB	WMDY	
1415	1396		*day				
1416	1397	6901	B6	6041	LDA	WMDY+2	
1417	1398	6904	84	1F	ANDA	#\$1F	
1418	1399	6906	BB	6040	ADDA	WMDY+1	
1419	1400	6909	B7	6040	STA	WMDY+1	
1420	1401	690C	39		RTS		
1421	1402						
1422	1403		*lower nibble of A,B to A+B				
1423	1404	690D		APLUSB	EQU	*	
1424	1405	690D	34	10	PSHS	X	
1425	1406	690F	34	04	PSHS	B	
1426	1407	6911	C6	0A	LDB	#10	
1427	1408	6913	3D		MUL		
1428	1409	6914	1F	01	TFR	D,X	
1429	1410	6916	35	04	PULS	B	
1430	1411	6918	3A		ABX		
1431	1412	6919	1F	10	TFR	X,D	
1432	1413	691B	1E	89	EXG	A,B	A:data
1433	1414	691D	35	90	PULS	X,PC	
1434	1415		*				
1435	1416		*-----				

1436	1417	691F		WRTFIL	EQU	*		
1437	1418			*calc file size				
1438	1419	691F	7D	604D		TST	MCFLG	0,FF(FE),else
1439	1420	6922	2B	08		BMI	WRTF1	
1440	1421	6924	FC	6061		LDD	MCAEP	
1441	1422	6927	B3	605F		SUBD	MCADP	
1442	1423	692A	20	04		BRA	WRTF2	
1443	1424	692C		WRTF1				
1444	1425	692C	DC	35		LDD	BASAEP	
1445	1426	692E	93	33		SUBD	BASADP	
1446	1427	6930	FD	6046	WRTF2	STD	FILSIZ	
1447	1428	6933	1F	02		TFR	D,Y	file size
1448	1429	6935	10BF	606B		STY	WK3	
1449	1430	6939	1E	89		EXG	A,B	
1450	1431	693B	8E	601F		LDX	#DIRENT	
1451	1432	693E	ED	88 1C		STD	\$1C,X	
1452	1433	6941	1E	89		EXG	A,B	
1453	1434							
1454	1435			*add header&footer				
1455	1436	6943	7D	604D		TST	MCFLG	0,FF(FE),else
1456	1437	6946	27	04		BEQ	WRT3	
1457	1438	6948	2B	07		BMI	WRT4	
1458	1439	694A	2A	08		BPL	WRT5	
1459	1440							
1460	1441	694C	C3	000A	WRT3	ADDD	#10	header&footer
1461	1442	694F	20	03		BRA	WRT5	
1462	1443	6951	C3	0003	WRT4	ADDD	#3	header
1463	1444	6954	1F	02	WRT5	TFR	D,Y	
1464	1445							
1465	1446			*calc counter (CNT:512byte units)				
1466	1447	6956	7F	604C		CLR	CNT	
1467	1448	6959	7C	604C	CALLOP	INC	CNT	
1468	1449	695C	83	0200		SUBD	#512	
1469	1450	695F	22	F8		BHI	CALLOP	
1470	1451	6961	B6	604C		LDA	CNT	
1471	1452	6964	B7	604E		STA	TPGFLG	set top page
1472	1453							
1473	1454	6967	7D	604D		TST	MCFLG	
1474	1455	696A	2B	05		BMI	WRTF3	

1475	1456	696C	BE	605F		LDX	MCADP	saved top adr
1476	1457	696F	20	02		BRA	WRTF4	
1477	1458							
1478	1459	6971	9E	33	WRTF3	LDX	BASADP	saved top adr
1479	1460	6973	BF	6067	WRTF4	STX	WK1	src topadr
1480	1461	6976	CE	5E00		LDU	#RWBUF	
1481	1462							
1482	1463							*get clusterL
1483	1464	6979	8E	601F		LDX	#DIRENT	
1484	1465	697C	EC	88 1A		LDD	\$1A,X	
1485	1466	697F	1E	89		EXG	A,B	B:clusterL
1486	1467							
1487	1468	6981	BE	601A		LDX	SCTCLT	sct no of top
1488	cluster							
1489	1469	6984	BF	6069		STX	WK2	
1490	1470	6987	BE	601C		LDX	SCTCLT+2	
1491	1471	698A	BF	606B		STX	WK2+2	
1492	1472							
1493	1473	698D	C0	02		SUBB	#2	for SD
1494	1474	698F	7D	6009		TST	SDHC	0:SD,1:SDHC
1495	1475	6992	27	01		BEQ	WR2	
1496	1476	6994	5A			DECB		
1497	1477	6995		WR2				
1498	1478							
1499	1479							*(X)..(X+3)+(B)x40h
1500	1480	6995	8E	6069		LDX	#WK2	
1501	1481	6998	17	01B5		LBSR	ADD40H	cal of sct no
1502	1482							
1503	1483							*set start sct no of file
1504	1484	699B	34	10		PSHS	X	
1505	1485	699D	8E	7022		LDX	#MSWTNG	
1506	1486	69A0	17	0383		LBSR	PSTRNG	
1507	1487	69A3	35	10		PULS	X	
1508	1488							
1509	1489	69A5		WRL0P				
1510	1490	69A5	86	2E		LDA	#'.	
1511	1491	69A7	BD	FBBB		JSR	OUTCHA	
1512	1492	69AA	10BE	606B		LDY	WK3	
1513	1493	69AE	1F	20		TFR	Y,D	

1514	1494	69B0	1083	0200		CMPD	#512	
1515	1495	69B4	23	04		BLS	WR3	
1516	1496	69B6	108E	0200		LDY	#512	
1517	1497	69BA		WR3				
1518	1498							
1519	1499					*write data to rwbuf		
1520	1500					* (X)->(U:RWBUF)		
1521	1501					* (Y):SIZE 512bytes or Less		
1522	1502	69BA	BE	6067		LDX	WK1	
1523	1503	69BD	CE	5E00		LDU	#RWBUF	
1524	1504	69C0	8D	44		BSR	WRSUB	
1525	1505							
1526	1506					*set to sctn0-3		
1527	1507	69C2	34	30		PSHS	X,Y	
1528	1508	69C4	8E	6069		LDX	#WK2	
1529	1509	69C7	108E	6004		LDY	#SCTN0	
1530	1510	69CB	C6	04		LDB	#4	
1531	1511	69CD	17	015D		LBSR	TRANS	
1532	1512	69D0	35	30		PULS	X,Y	
1533	1513							
1534	1514					*write data to sd		
1535	1515	69D2	17	008F		LBSR	WRTSD	...not wrtsd1
1536	1516							
1537	1517					*inc sctn0-3		
1538	1518	69D5	8E	6069		LDX	#WK2	
1539	1519	69D8	17	0166		LBSR	INCX4	
1540	1520	69DB	BE	6067		LDX	WK1	src topadr
1541	1521	69DE	3089	0200		LEAX	512,X	
1542	1522					*check top page		
1543	1523	69E2	7D	604E		TST	TPGFLG	top page?
1544	1524	69E5	27	0D		BEQ	WR6	
1545	1525	69E7	7D	604D		TST	MCFLG	
1546	1526	69EA	27	04		BEQ	WR4	
1547	1527	69EC	2B	04		BMI	WR5	
1548	1528	69EE	2A	04		BPL	WR6	
1549	1529	69F0	30	1E	WR4	LEAX	-2,X	rewind 3or5
1550	1530	69F2	30	1D	WR5	LEAX	-3,X	
1551	1531	69F4		WR6				
1552	1532	69F4	7F	604E		CLR	TPGFLG	

1553	1533	69F7	BF	6067	STX	WK1	
1554	1534	69FA	7A	604C	DEC	CNT	
1555	1535		*	BEQ WREND			
1556	1536		*	BRA WRLOP			
1557	1537	69FD	2A	A6	BPL	WRLOP	
1558	1538						
1559	1539	69FF	8E	703B	WREND	LDX	#MSWTED
1560	1540	6A02	17	0321		LBSR	PSTRNG
1561	1541	6A05	39		RTS		
1562	1542						
1563	1543		*	-----			
1564	1544		*	write data to rwbuf			
1565	1545		*	(X)->(U:RWBUF)			
1566	1546		*	(Y):SIZE 512bytes or Less			
1567	1547		*	clear rwbuf			
1568	1548	6A06		WRSUB EQU *			
1569	1549	6A06	34	30	PSHS	X,Y	
1570	1550	6A08	4F		CLRA		
1571	1551	6A09	8E	5E00	LDX	#RWBUF	
1572	1552	6A0C	108E	0200	LDY	#512	
1573	1553	6A10	A7	80	WWCLOP	STA	,X+
1574	1554	6A12	31	3F	LEAY	-1,Y	
1575	1555	6A14	26	FA	BNE	WWCLOP	
1576	1556	6A16	35	30	PULS	X,Y	
1577	1557						
1578	1558		*	set data to rwbuf			
1579	1559	6A18	CE	5E00	LDU	#RWBUF	
1580	1560						
1581	1561	6A1B	7D	604E	TST	TPGFLG	top page?
1582	1562	6A1E	27	24	BEQ	WRLOP1	
1583	1563						
1584	1564	6A20	7D	604D	TST	MCFLG	0,\$FF(\$FE),etc
1585	1565	6A23	2B	15	BMI	WRS1	
1586	1566	6A25		WRS0			
1587	1567		*	add header of machine code			
1588	1568	6A25	34	10	PSHS	X	
1589	1569	6A27	4F		CLRA		
1590	1570	6A28	A7	C0	STA	,U+	00
1591	1571	6A2A	BE	6046	LDX	FILSIZ	

1592	1572	6A2D	AF	C1	STX	,U++	file size
1593	1573	6A2F	BE	605F	LDX	MCADP	
1594	1574	6A32	AF	C1	STX	,U++	save adr
1595	1575	6A34	31	3B	LEAY	-5,Y	
1596	1576	6A36	35	10	PULS	X	
1597	1577	6A38	20	0A	BRA	WRL0P1	
1598	1578						
1599	1579	6A3A		WRS1			
1600	1580				* add header of basic file		
1601	1581	6A3A	86	FF	LDA	#\$FF	
1602	1582	6A3C	A7	C0	STA	,U+	
1603	1583	6A3E	A7	C0	STA	,U+	
1604	1584	6A40	A7	C0	STA	,U+	
1605	1585	6A42	31	3D	LEAY	-3,Y	
1606	1586						
1607	1587	6A44	A6	80	WRL0P1 LDA	,X+	
1608	1588	6A46	A7	C0	STA	,U+	
1609	1589	6A48	31	3F	LEAY	-1,Y	
1610	1590	6A4A	26	F8	BNE	WRL0P1	
1611	1591						
1612	1592				*check last page		
1613	1593	6A4C	B6	604C	LDA	CNT	
1614	1594	6A4F	81	01	CMPA	#1	
1615	1595	6A51	26	10	BNE	WRS2	
1616	1596				*add footer		
1617	1597	6A53	86	FF	LDA	#\$FF	
1618	1598	6A55	A7	C0	STA	,U+	
1619	1599	6A57	4F		CLRA		
1620	1600	6A58	A7	C0	STA	,U+	
1621	1601	6A5A	A7	C0	STA	,U+	
1622	1602	6A5C	BE	6063	LDX	MCEXP	
1623	1603	6A5F	AF	C1	STX	,U++	
1624	1604	6A61	31	3B	LEAY	-5,Y	
1625	1605	6A63	39	WRS2	RTS		
1626	1606						
1627	1607				*write data to sd		
1628	1608	6A64		WRTSD EQU	*		
1629	1609	6A64	17	01A0	LBSR	SETSCT	*set SCT No
1630	1610				*copy sct no to cmd param		

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1631 1611 6A67 8E 6004 WRTSD1 LDX #SCTN0
1632 1612 6A6A 108E 6E5A LDY #CMD24+1
1633 1613 6A6E C6 04 LDB #4
1634 1614 6A70 17 00BA LBSR TRANS
1635 1615 *send cmd
1636 1616 6A73 17 01C0 LBSR SETCKL
1637 1617 6A76 C6 06 LDB #6
1638 1618 6A78 8E 6E59 LDX #CMD24
1639 1619 6A7B A6 80 SDW1 LDA ,X+
1640 1620 6A7D 17 0149 LBSR SNDDAT
1641 1621 6A80 5A DECB
1642 1622 6A81 26 F8 BNE SDW1
1643 1623 *get cmd response
1644 1624 6A83 SDW2
1645 1625 6A83 17 015F LBSR GETDAT
1646 1626 6A86 B6 6008 LDA RESDAT
1647 1627 6A89 81 00 CMPA #$00
1648 1628 6A8B 26 F6 BNE SDW2
1649 1629 *1byte skip
1650 1630 6A8D 86 FF LDA #$FF
1651 1631 6A8F 17 0137 LBSR SNDDAT 1byte skip
1652 1632
1653 1633 *send data packet
1654 1634 6A92 8E 5E00 LDX #RWBUF <-temp -1=""
1655 008c="" 0093="" 00ab="" 00b3="" 00c3="" 011a="" 011d="" 0126=""
1656 0129="" 012b="" 0133="" 01="" 0200="" 02="" 03="" 04="" 05="" 06=""
1657 08="" 0:sd="" 0f="" 108e="" 10be="" 12="" 1635="" 1636="" 1637=""
1658 1638="" 1639="" 1640="" 1641="" 1642="" 1643="" 1644="" 1645=""
1659 1646="" 1647="" 1648="" 1649="" 1650="" 1651="" 1652="" 1653=""
1660 1654="" 1655="" 1656="" 1657="" 1658="" 1659="" 1660="" 1661=""
1661 1662="" 1663="" 1664="" 1665="" 1666="" 1667="" 1668="" 1669=""
1662 1670="" 1671="" 1672="" 1673="" 1674="" 1675="" 1676="" 1677=""
1663 1678="" 1679="" 1680="" 1681="" 1682="" 1683="" 1684="" 1685=""
1664 1686="" 1687="" 1688="" 1689="" 1690="" 1691="" 1692="" 1693=""
1665 1694="" 1695="" 1696="" 1697="" 1698="" 1699="" 16="" 1700="" 1701=""
1666 1702="" 1703="" 1704="" 1705="" 1706="" 1707="" 1708="" 1709=""
1667 1710="" 1711="" 1712="" 1713="" 1714="" 1715="" 1716="" 1717=""
1668 1718="" 1719="" 1720="" 1721="" 1722="" 1723="" 1724="" 1725=""
1669 1726="" 1727="" 1728="" 1729="" 1730="" 1731="" 1732="" 1733=""

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1670 1734="" 1735="" 1736="" 1737="" 1738="" 1739="" 1740="" 1741=""
1671 1742="" 1743="" 1744="" 1745="" 1746="" 1747="" 1748="" 1749=""
1672 1750="" 1751="" 1752="" 1753="" 1754="" 1755="" 1756="" 1757=""
1673 1758="" 1759="" 1760="" 1761="" 1762="" 1763="" 1764="" 1765=""
1674 1766="" 1767="" 1768="" 1769="" 1770="" 1771="" 1772="" 1773=""
1675 1774="" 1775="" 1776="" 1777="" 1778="" 1779="" 1780="" 1781=""
1676 1782="" 1783="" 1784="" 1785="" 1786="" 1787="" 1788="" 1789=""
1677 1790="" 1791="" 1792="" 1793="" 1794="" 1795="" 1796="" 1797=""
1678 1798="" 1799="" 17="" 1800="" 1801="" 1802="" 1803="" 1804="" 1805=""
1679 1806="" 1807="" 1808="" 1809="" 1810="" 1811="" 1812="" 1813=""
1680 1814="" 1815="" 1816="" 1817="" 1818="" 1819="" 1820="" 1821=""
1681 1822="" 1823="" 1824="" 1825="" 1826="" 1827="" 1828="" 1829=""
1682 1830="" 1831="" 1832="" 1833="" 1834="" 1835="" 1836="" 1837=""
1683 1838="" 1839="" 1840="" 1841="" 1842="" 1843="" 1844="" 1845=""
1684 1846="" 1847="" 1848="" 1849="" 1850="" 1851="" 1852="" 1853=""
1685 1854="" 1855="" 1856="" 1857="" 1858="" 1859="" 1="" 1a="" 1e=""
1686 1f="" 20="" 24="" 26="" 27="" 2="" 31="" 34="" 35="" 36="" 39="" 3=""
1687 3a="" 3d="" 3f="" 40="" 44="" 48="" 4b="" 4f="" 54="" 5a="" 5b=""
1688 5e00="" 5e="" 5f="" 6004="" 6008="" 6009="" 600e="" 6012="" 6016=""
1689 601f="" 605a="" 605b="" 68="" 6a95="" 6a99="" 6a9b="" 6a9e="" 6aa0=""
1690 6aa3="" 6aa5="" 6aa7="" 6aa9="" 6aac="" 6aaf="" 6ab2="" 6ab5=""
1691 6ab7="" 6ab9="" 6abc="" 6abf="" 6ac1="" 6ac3="" 6ac4="" 6ac7=""
1692 6acb="" 6acd="" 6acf="" 6ad2="" 6ad5="" 6ad7="" 6ad9="" 6ada=""
1693 6adc="" 6adf="" 6ae2="" 6ae5="" 6ae9="" 6aeb="" 6aed="" 6af0=""
1694 6af1="" 6af4="" 6af7="" 6af9="" 6afb="" 6afd="" 6b00="" 6b03=""
1695 6b05="" 6b07="" 6b09="" 6b0c="" 6b0e="" 6b10="" 6b13="" 6b14=""
1696 6b16="" 6b18="" 6b1a="" 6b1c="" 6b1e="" 6b21="" 6b23="" 6b25=""
1697 6b27="" 6b29="" 6b2c="" 6b2d="" 6b2f="" 6b31="" 6b33="" 6b34=""
1698 6b36="" 6b38="" 6b3a="" 6b3c="" 6b3d="" 6b3f="" 6b41="" 6b43=""
1699 6b45="" 6b47="" 6b49="" 6b4b="" 6b4d="" 6b4f="" 6b50="" 6b52=""
1700 6b54="" 6b56="" 6b58="" 6b5a="" 6b5c="" 6b5e="" 6b60="" 6b62=""
1701 6b64="" 6b66="" 6b68="" 6b6a="" 6b6c="" 6b6e="" 6b70="" 6b71=""
1702 6b73="" 6b75="" 6b78="" 6b7c="" 6b7e="" 6b80="" 6b83="" 6b85=""
1703 6b88="" 6b8a="" 6b8c="" 6b8d="" 6b8f="" 6b91="" 6b94="" 6b96=""
1704 6b98="" 6b9b="" 6b9f="" 6ba1="" 6ba4="" 6ba6="" 6ba8="" 6baa=""
1705 6bab="" 6bae="" 6bb1="" 6bb5="" 6bb7="" 6bb9="" 6bbc="" 6bbe=""
1706 6bc0="" 6bc1="" 6bc3="" 6bc6="" 6bc8="" 6bc9="" 6bcb="" 6bce=""
1707 6bd0="" 6bd2="" 6bd4="" 6bd6="" 6bd9="" 6bdb="" 6bdd="" 6bde=""
1708 6be0="" 6be1="" 6be3="" 6be5="" 6be7="" 6be9="" 6beb="" 6bec=""

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1709 6bed="" 6bee="" 6bf0="" 6bf3="" 6bf4="" 6c="" 6e53="" 6e54="" 70=""
1710 7d="" 7f="" 80="" 81="" 84="" 86="" 88="" 89="" 8a="" 8b="" 8d=""
1711 8e="" 96="" a0="" a6="" a7="" a:fill="" a="" abx="" ad="" add0=""
1712 add1="" add2="" add3="" add40h="" adda="" addb="" adr="" adx0=""
1713 adx1="" adx2="" adx3="" anda="" b1-="" b6="" b7="" b:cnt="" bcc=""
1714 beq="" bne="" bra="" bsr="" buf="" busy="" c6="" cb="" ckrs="" clr=""
1715 cluster="" cmd="" cmpa="" copy="" crch="" crcl="" cs:b0="L" d=""
1716 data="" decb="" df="" dir="" dirofs="" disp="" dpgno="" dspb1=""
1717 dspb2="" dspb3="" dspbin="" dspbuf="" e5="" ec="" ed="" equ="" exg=""
1718 f4="" f5="" f6="" f7="" f8b2="" f8d6="" f9="" fb="" fc="" fd9e=""
1719 fe="" ff3b="" ff77="" ff="" fill="" filop="" get="" getd1=""
1720 getdat="" inc="" incx4="" lbr="" lda="" ldb="" ldd="" ldx="" ldy=""
1721 leay="" lsla="" lsra="" miso:b1="H(for" no="" nop="" of="" ora=""
1722 param="" pbdatt="" pcrLf="" pia="" pshs="" puls="" rd1="" rd1sct=""
1723 rd2="" rd3="" read="" resdat="" response="" return="" rts="" save=""
1724 sct="" sd:2="" sdhc:4="" sdhc="" send="" setckh="" setckl=""
1725 setsno="" snddl="" snddat="" snddt1="" src="" sta="" temp="" tfr=""
1726 to="" token="" trans="" trlop="" tst="" until="" wait="" wrdlop=""
1727 wrdskp="" wrf1="" wrf2="" wrf3="" wrtdir="" wrtfat="" wrtsd1=""
1728 wspfl0="" x40h="" y:dst="" y="">C
1729 1860 6BF5 79 6008          ROL      RESDAT      C->b0
1730 1861 6BF8 5A              DECB
1731 1862 6BF9 26 EE          BNE      GETD1
1732 1863 6BFB 8D 39          BSR      SETCKL
1733 1864 6BFD 35 86          PULS     A,B,PC
1734 1865                    *
1735 1866                    *-----
1736 1867                    *input SCT No
1737 1868 6BFF          INSCT  EQU      *
1738 1869 6BFF 8E 6F5E          LDX      #MSSCT
1739 1870 6C02 17 0121          LBSR     PSTRNG
1740 1871 6C05 8D 75          BSR      GET8CH      get 8chr, convert
1741 4bin
1742 1872
1743 1873 6C07          SETSCT  EQU      *
1744 1874 6C07 34 32          PSHS     A,X,Y
1745 1875 6C09 7D 6009          TST      SDHC      0:SD,1:SDHC
1746 1876 6C0C 26 1A          BNE      CALCHC
1747 1877                    *SD card

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1748	1878	6C0E	8E	6005	LDX	#SCTN1	
1749	1879	6C11	108E	6004	LDY	#SCTN0	
1750	1880	6C15	C6	03	LDB	#3	
1751	1881	6C17	17	FF13	LBSR	TRANS	
1752	1882	6C1A	7F	6007	CLR	SCTN3	
1753	1883				*calc top adrs of sct		
1754	1884	6C1D		CALCSD	EQU	*	
1755	1885	6C1D	78	6007	LSL	SCTN3	
1756	1886	6C20	79	6006	ROL	SCTN2	
1757	1887	6C23	79	6005	ROL	SCTN1	
1758	1888	6C26	35	B2	PULS	A,X,Y,PC	
1759	1889						
1760	1890				*SDHC card		
1761	1891				*calc top adrs of sct		
1762	1892	6C28		CALCHC	EQU	*	
1763	1893	6C28	8E	6004	LDX	#SCTN0	
1764	1894	6C2B	108E	6E54	LDY	#CMD17+1	
1765	1895	6C2F	C6	04	LDB	#4	
1766	1896	6C31	17	FEF9	LBSR	TRANS	
1767	1897	6C34	35	B2	PULS	A,X,Y,PC	
1768	1898						
1769	1899				*		
1770	1900				*set sck:L		
1771	1901	6C36		SETCKL	EQU	*	
1772	1902	6C36	34	02	PSHS	A	
1773	1903	6C38	86	34	LDA	#\$34	ca2=L
1774	1904	6C3A	B7	FD9F	STA	PBCTL	
1775	1905	6C3D	35	82	PULS	A,PC	
1776	1906						
1777	1907				*set sck:H		
1778	1908	6C3F		SETCKH	EQU	*	
1779	1909	6C3F	34	02	PSHS	A	
1780	1910	6C41	86	3C	LDA	#\$3C	ca2=H
1781	1911	6C43	B7	FD9F	STA	PBCTL	
1782	1912	6C46	35	82	PULS	A,PC	
1783	1913						
1784	1914				*binary(A)->ascii(A)(B)		
1785	1915	6C48		DSPBIN	EQU	*	
1786	1916	6C48	34	06	PSHS	A,B	

1787	1917	6C4A	1F	89		TFR	A,B
1788	1918	6C4C	44			LSRA	
1789	1919	6C4D	44			LSRA	
1790	1920	6C4E	44			LSRA	
1791	1921	6C4F	44			LSRA	
1792	1922	6C50	8A	30		ORA	#\$30
1793	1923	6C52	C4	0F		ANDB	#\$0F
1794	1924	6C54	CA	30		ORB	#\$30
1795	1925	6C56	81	39		CMPA	#\$39
1796	1926	6C58	23	02		BLS	BI21
1797	1927	6C5A	8B	07		ADDA	#\$07
1798	1928	6C5C	C1	39	BI21	CMPB	#\$39
1799	1929	6C5E	23	02		BLS	BI22
1800	1930	6C60	CB	07		ADDB	#\$07
1801	1931	6C62	BD	FBBB	BI22	JSR	OUTCHA
1802	1932	6C65	1F	98		TFR	B,A
1803	1933	6C67	BD	FBBB		JSR	OUTCHA
1804	1934	6C6A	86	20		LDA	#\$20
1805	1935	6C6C	BD	FBBB		JSR	OUTCHA
1806	1936	6C6F	35	86		PULS	A,B,PC
1807	1937						
1808	1938		*crlf				
1809	1939	6C71		PCRLF	EQU	*	
1810	1940	6C71	86	0D		LDA	#CR
1811	1941	6C73	BD	FBBB		JSR	OUTCHA
1812	1942	6C76	86	0A		LDA	#LF
1813	1943	6C78	BD	FBBB		JSR	OUTCHA
1814	1944	6C7B	39			RTS	
1815	1945						
1816	1946		*				
1817	1947	6C7C		GET8CH	EQU	*	
1818	1948		*input chr				
1819	1949	6C7C	8E	606F		LDX	#INPBUF
1820	1950	6C7F	5F			CLRB	
1821	1951	6C80	BD	6E2D	CLOP	JSR	INOUTC
1822	1952	6C83	81	0D		CMPA	#CR
1823	1953	6C85	27	07		BEQ	INEND
1824	1954	6C87	A7	85		STA	B,X
1825	1955	6C89	5C			INCB	

1826	1956	6C8A	C1	08		CMPB	#8
1827	1957	6C8C	25	F2		BLO	CLOP
1828	1958						
1829	1959	6C8E			INEND		
1830	1960				*shift		
1831	1961	6C8E	5D			TSTB	
1832	1962	6C8F	27	21		BEQ	CLEND
1833	1963	6C91	8E	606F		LDX	#INPBUF
1834	1964	6C94	3A			ABX	
1835	1965	6C95	108E	6077		LDY	#INPBUF+8
1836	1966	6C99	34	04		PSHS	B
1837	1967	6C9B	A6	82	SFLOP	LDA	, -X
1838	1968	6C9D	A7	A2		STA	, -Y
1839	1969	6C9F	5A			DECB	
1840	1970	6CA0	26	F9		BNE	SFLOP
1841	1971				*clr		
1842	1972	6CA2	C6	08	CL1	LDB	#8
1843	1973	6CA4	E0	E0		SUBB	, S+
1844	1974	6CA6	27	0A		BEQ	CLEND
1845	1975	6CA8	8E	606F		LDX	#INPBUF
1846	1976	6CAB	86	30		LDA	#'0
1847	1977	6CAD	A7	80	CLLOP	STA	, X+
1848	1978	6CAF	5A			DECB	
1849	1979	6CB0	26	FB		BNE	CLLOP
1850	1980	6CB2			CLEND		
1851	1981						
1852	1982				*convert to bin		
1853	1983	6CB2	C6	04		LDB	#4
1854	1984	6CB4	8E	606F		LDX	#INPBUF
1855	1985	6CB7	108E	6004		LDY	#SCTN0
1856	1986	6CBB			CVT1		
1857	1987	6CBB	A6	80		LDA	, X+
1858	1988	6CBD	8D	1E		BSR	CVBIN
1859	1989	6CBF	25	16		BCS	EREND1
1860	1990	6CC1	48			LSLA	
1861	1991	6CC2	48			LSLA	
1862	1992	6CC3	48			LSLA	
1863	1993	6CC4	48			LSLA	
1864	1994	6CC5	34	02		PSHS	A

1865	1995	6CC7	A6	80		LDA	,X+	
1866	1996	6CC9	8D	12		BSR	CVBIN	
1867	1997	6CCB	25	08		BCS	EREND0	
1868	1998	6CCD	AB	E0		ADDA	,S+	
1869	1999	6CCF	A7	A0		STA	,Y+	
1870	2000	6CD1	5A			DECB		
1871	2001	6CD2	26	E7		BNE	CVT1	
1872	2002	6CD4	39			RTS		
1873	2003							
1874	2004	6CD5	35	02	EREND0	PULS	A	
1875	2005	6CD7	8E	6FA4	EREND1	LDX	#MSERR	
1876	2006	6CDA	8D	4A		BSR	PSTRNG	
1877	2007	6CDC	39		EEND	RTS		
1878	2008							
1879	2009				*-----			
1880	2010				*ascii -> bin			
1881	2011	6CDD			CVBIN	EQU	*	
1882	2012	6CDD	80	30		SUBA	#'0	
1883	2013	6CDF	2B	11		BMI	CVERR	
1884	2014	6CE1	81	09		CMPA	#9	
1885	2015	6CE3	23	0A		BLS	CVBEND	
1886	2016	6CE5	81	11		CMPA	#\$11	
1887	2017	6CE7	25	09		BLO	CVERR	
1888	2018	6CE9	81	16		CMPA	#\$16	
1889	2019	6CEB	22	05		BHI	CVERR	
1890	2020	6CED	80	07		SUBA	#7	
1891	2021	6CEF			CVBEND			
1892	2022	6CEF	1C	FE		ANDCC	#\$FE	reset C
1893	2023	6CF1	39			RTS		
1894	2024	6CF2	1A	01	CVERR	ORCC	#\$01	set C
1895	2025	6CF4	39			RTS		
1896	2026							
1897	2027				*-----			
1898	2028				*add 2bytes to 4bytes			
1899	2029				* X,X+1,X+2,X+3 + Y,Y+1			
1900	2030				* data X is big endian			
1901	2031				* data Y is little endian			
1902	2032	6CF5			ADD42	EQU	*	
1903	2033	6CF5	34	36		PSHS	A,B,X,Y	

1904	2034	6CF7	32	7C		LEAS	-4,S
1905	2035	6CF9	C6	04		LDB	#4
1906	2036	6CFB	A6	80	AD1	LDA	,X+
1907	2037	6CFD	A7	E0		STA	,S+
1908	2038	6CFF	5A			DECB	
1909	2039	6D00	26	F9		BNE	AD1
1910	2040						
1911	2041				*add (lower)		
1912	2042	6D02	A6	A0		LDA	,Y+
1913	2043	6D04	AB	E2		ADDA	, -S
1914	2044	6D06	A7	E4		STA	,S
1915	2045	6D08	24	02		BCC	AD2
1916	2046	6D0A	6C	7F		INC	-1,S
1917	2047				*add (upper)		
1918	2048	6D0C	A6	A4	AD2	LDA	,Y
1919	2049	6D0E	AB	E2		ADDA	, -S
1920	2050	6D10	A7	E4		STA	,S
1921	2051	6D12	24	02		BCC	AD3
1922	2052	6D14	6C	7F		INC	-1,S
1923	2053	6D16	30	1C	AD3	LEAX	-4,X
1924	2054	6D18	32	7E		LEAS	-2,S
1925	2055	6D1A	C6	04		LDB	#4
1926	2056	6D1C	A6	E0	AD4	LDA	,S+
1927	2057	6D1E	A7	80		STA	,X+
1928	2058	6D20	5A			DECB	
1929	2059	6D21	26	F9		BNE	AD4
1930	2060	6D23	35	36		PULS	A,B,X,Y
1931	2061	6D25	39			RTS	
1932	2062						
1933	2063				*-----		
1934	2064				* pstring		
1935	2065	6D26			PSTRNG EQU *		
1936	2066	6D26	34	02		PSHS	A
1937	2067	6D28	A6	80	PSLOP	LDA	,X+
1938	2068	6D2A	81	04		CMPA	#4
1939	2069	6D2C	27	05		BEQ	PSEND
1940	2070	6D2E	BD	FBBB		JSR	OUTCHA
1941	2071	6D31	20	F5		BRA	PSLOP
1942	2072	6D33	35	02	PSEND	PULS	A

1943	2073	6D35	39		RTS			
1944	2074							
1945	2075				*-----			
1946	2076				*disp MM-DD-YY			
1947	2077				*indata A(Low),B(High)			
1948	2078	6D36		DSPYMD	EQU	*		
1949	2079	6D36	1E	89	EXG		A,B	
1950	2080	6D38	34	10	PSHS		X	
1951	2081	6D3A	34	06	PSHS		A,B	
1952	2082	6D3C	C4	E0	ANDB		#\$E0	
1953	2083	6D3E	54		LSRB			
1954	2084	6D3F	54		LSRB			
1955	2085	6D40	54		LSRB			
1956	2086	6D41	54		LSRB			
1957	2087	6D42	54		LSRB			
1958	2088	6D43	47		ASRA			
1959	2089	6D44	24	02	BCC		DYMD1	
1960	2090	6D46	CB	08	ADDB		#8	
1961	2091	6D48	F7	603F	DYMD1	STB	WMDY	MM
1962	2092	6D4B	35	06	PULS		A,B	
1963	2093	6D4D	34	04	PSHS		B	
1964	2094	6D4F	C4	1F	ANDB		#\$1F	
1965	2095	6D51	F7	6040	STB		WMDY+1	DD
1966	2096	6D54	35	04	PULS		B	
1967	2097							
1968	2098	6D56	44		LSRA			
1969	2099				* ASLB			
1970	2100				* BCC DYMD2			
1971	2101				* INCA			
1972	2102	6D57		DYMD2				
1973	2103	6D57	80	14	SUBA	#20		A+1980-2000
1974	2104	6D59	B7	6041	STA	WMDY+2		YY
1975	2105							
1976	2106				*disp MM-DD-YY			
1977	2107	6D5C	86	20	LDA	#SPC		
1978	2108	6D5E	BD	FBBB	JSR	OUTCHA		
1979	2109	6D61	8E	603F	LDX	#WMDY		
1980	2110	6D64	A6	80	LDA	,X+		
1981	2111	6D66	8D	14	BSR	DB2ASC		MM

1982	2112	6D68	86	2D	LDA	#' -	
1983	2113	6D6A	BD	FBBB	JSR	OUTCHA	
1984	2114	6D6D	A6	80	LDA	,X+	
1985	2115	6D6F	8D	0B	BSR	DB2ASC	DD
1986	2116	6D71	86	2D	LDA	#' -	
1987	2117	6D73	BD	FBBB	JSR	OUTCHA	
1988	2118	6D76	A6	80	LDA	,X+	
1989	2119	6D78	8D	02	BSR	DB2ASC	YY
1990	2120	6D7A	35	90	PULS	X,PC	
1991	2121		*				
1992	2122		*-----				
1993	2123		*convert decimal value to ascii code				
1994	2124		*(A:BIN)->ASCII				
1995	2125	6D7C		DB2ASC EQU	*		
1996	2126	6D7C	34	06	PSHS	A,B	
1997	2127	6D7E	1F	89	TFR	A,B	
1998	2128	6D80	86	30	LDA	#\$30	
1999	2129	6D82	C1	0A	DBLOP	CMPB	#10
2000	2130	6D84	25	05	BLO	DB2	
2001	2131	6D86	C0	0A	SUBB	#10	
2002	2132	6D88	4C		INCA		
2003	2133	6D89	20	F7	BRA	DBLOP	
2004	2134	6D8B	BD	FBBB	DB2	JSR	OUTCHA
2005	2135	6D8E	1F	98	TFR	B,A	
2006	2136	6D90	8B	30	ADDA	#\$30	
2007	2137	6D92	BD	FBBB	JSR	OUTCHA	
2008	2138	6D95	35	86	PULS	A,B,PC	
2009	2139						
2010	2140		*disp FILE size				
2011	2141	6D97		DSPSIZ EQU	*		
2012	2142	6D97	34	10	PSHS	X	
2013	2143	6D99	BE	6042	LDX	WFSIZ	
2014	2144	6D9C	27	07	BEQ	DSS1	if 64k over?
2015	2145	6D9E	8E	706F	LDX	#MSOVR	
2016	2146	6DA1	8D	83	BSR	PSTRNG	
2017	2147	6DA3	20	40	BRA	DSEND	
2018	2148						
2019	2149	6DA5		DSS1			
2020	2150		*	LDA #SPC			

2021	2151		* JSR OUTCHA					
2022	2152		* /1024(400h)					
2023	2153	6DA5	B6	6044		LDA	WFSIZ+2	under 64k
2024	2154	6DA8	44			LSRA		
2025	2155	6DA9	44			LSRA		
2026	2156		*disp dec					
2027	2157	6DAA	1F	89		TFR	A,B	
2028	2158	6DAC	81	64		CMPA	#100	
2029	2159	6DAE	23	10		BLS	DSS13	
2030	2160	6DB0	86	30		LDA	#'0	
2031	2161	6DB2	C1	64	DSL0P1	CMPB	#100	
2032	2162	6DB4	25	05		BLO	DSS12	
2033	2163	6DB6	C0	64		SUBB	#100	
2034	2164	6DB8	4C			INCA		
2035	2165	6DB9	20	F7		BRA	DSL0P1	
2036	2166	6DBB	BD	FBBB	DSS12	JSR	OUTCHA	
2037	2167	6DBE	20	0B		BRA	DSS14	
2038	2168							
2039	2169		*under 100					
2040	2170	6DC0	86	20	DSS13	LDA	#SPC	
2041	2171	6DC2	BD	FBBB		JSR	OUTCHA	
2042	2172	6DC5	86	20		LDA	#SPC	
2043	2173	6DC7	C1	0A		CMPB	#10	
2044	2174	6DC9	25	0B		BLO	DSS2	
2045	2175							
2046	2176	6DCB	86	30	DSS14	LDA	#'0	
2047	2177	6DCD	C1	0A	DSL0P2	CMPB	#10	
2048	2178	6DCF	25	05		BLO	DSS2	
2049	2179	6DD1	C0	0A		SUBB	#10	
2050	2180	6DD3	4C			INCA		
2051	2181	6DD4	20	F7		BRA	DSL0P2	
2052	2182							
2053	2183	6DD6	BD	FBBB	DSS2	JSR	OUTCHA	10's place
2054	2184		*under 10					
2055	2185	6DD9	1F	98	DSS21	TFR	B,A	
2056	2186	6ddb	8B	31		ADDA	#\$31	'1
2057	2187	6DDD	BD	FBBB	DSS3	JSR	OUTCHA	1's place
2058	2188	6DE0	86	6B		LDA	#'k	
2059	2189	6DE2	BD	FBBB		JSR	OUTCHA	

2060	2190	6DE5	86	20	DSEND	LDA	#SPC
2061	2191	6DE7	BD	FB	BB	JSR	OUTCHA
2062	2192	6DEA	35	90		PULS	X,PC
2063	2193						
2064	2194				*-----		
2065	2195				*disp cluster		
2066	2196				*return:clusterH,spc,clusterL		
2067	2197	6DEC			DSPCLS EQU *		
2068	2198	6DEC	34	10		PSHS	X
2069	2199	6DEE	86	20		LDA	#SPC
2070	2200	6DF0	BD	FB	BB	JSR	OUTCHA
2071	2201	6DF3	8E	60	48	LDX	#CLSTH
2072	2202	6DF6	C6	02		LDB	#2
2073	2203	6DF8	A6	80	DCL0P1	LDA	,X+
2074	2204	6DFA	17	FF	7F	LBSR	DB2ASC
2075	2205	6DFD	5A			DECB	
2076	2206	6DFE	26	F8		BNE	DCL0P1
2077	2207	6E00	86	20		LDA	#SPC
2078	2208	6E02	BD	FB	BB	JSR	OUTCHA
2079	2209	6E05	C6	02		LDB	#2
2080	2210	6E07	A6	80	DCL0P2	LDA	,X+
2081	2211	6E09	17	FF	70	LBSR	DB2ASC
2082	2212	6E0C	5A			DECB	
2083	2213	6E0D	26	F8		BNE	DCL0P2
2084	2214	6E0F	35	90		PULS	X,PC
2085	2215						
2086	2216				*-----		
2087	2217	6E11			FNMIN EQU *		
2088	2218				*clr buf		
2089	2219	6E11	8E	60	5A	LDX	#FNAME+11
2090	2220	6E14	C6	0B		LDB	#8+3
2091	2221	6E16	86	20		LDA	#SPC
2092	2222	6E18	A7	82	FNIN0	STA	, -X
2093	2223	6E1A	5A			DECB	
2094	2224	6E1B	26	FB		BNE	FNIN0
2095	2225						
2096	2226				*fname in		
2097	2227	6E1D	34	24		PSHS	B,Y
2098	2228	6E1F	10	8E	60	4F	LDY #FNAME

```

2099 2229 6E23 C6 08          LDB      #8
2100 2230 6E25 8D 0D          BSR      LININ
2101 2231 6E27 86 20          LDA      #$20
2102 2232 6E29 A7 3F          STA      -1,Y
2103 2233 6E2B 35 A4          PULS     B,Y,PC
2104 2234
2105 2235          *-----
2106 2236          *
2107 2237 6E2D BD FBB8 INOUTC JSR      INCHA
2108 2238 6E30 BD FBFB          JSR      OUTCHA
2109 2239 6E33 39          RTS
2110 2240
2111 2241          *-----
2112 2242          * line input
2113 2243          * input: B:count, Y:buf
2114 2244          * output: FILBUF (terminate $00)
2115 2245          * temp buf $043D - $053C
2116 2246 6E34          LININ EQU      *
2117 2247 6E34 0F BF          CLR      $BF
2118 2248 6E36 BD D807          JSR      LININP
2119 2249
2120 2250 6E39 8E 043D          LDX      #$043D
2121 2251 6E3C A6 80 LINLOP LDA      ,X+
2122 2252 6E3E A7 A0          STA      ,Y+
2123 2253 6E40 5A          DECB
2124 2254 6E41 27 03          BEQ      LEND
2125 2255 6E43 4D          TSTA
2126 2256 6E44 26 F6          BNE      LINLOP
2127 2257 6E46 39 LEND RTS
2128 2258
2129 2259          *
2130 2260          *****
2131 2261          *===== parameter =====
2132 2262 6E47 40 CMD0 FCB $40,$00,$00,$00,$00,$95 '@
2133 2263 6E4D 41 CMD1 FCB $41,$00,$00,$00,$00,$FF 'A
2134 2264 6E53 51 CMD17 FCB $51,$00,$00,$00,$00,$FF 'Q
2135 2265 6E59 58 CMD24 FCB $58,$00,$00,$00,$00,$FF 'X
2136 2266 6E5F 48 CMD8 FCB $48,$00,$00,$01,$AA,$87 'H
2137 2267 6E65 7A CMD58 FCB $7A,$00,$00,$00,$00,$FF 'z

```

2138	2268	6E6B	77	CMD55	FCB	\$77,\$00,\$00,\$00,\$00,\$FF 'w
2139	2269	6E71	69	ACMD41	FCB	\$69,\$40,\$FF,\$80,\$00,\$FF 'i
2140	2270	6E77	49	CMD9	FCB	\$49,\$00,\$00,\$00,\$00,\$FF 'I
2141	2271		*			
2142	2272	6E7D	43	MCMD0	FCC	/CMD0/,CR,LF
2143	2273	6E83	04		FCB	4
2144	2274	6E84	43	MCMD8	FCC	/CMD8/,CR,LF
2145	2275	6E8A	04		FCB	4
2146	2276	6E8B	43	MCMD58	FCC	/CMD58/,CR,LF
2147	2277	6E92	04		FCB	4
2148	2278	6E93	43	MCMD55	FCC	/CMD55/,CR,LF
2149	2279	6E9A	04		FCB	4
2150	2280	6E9B	41	MCMD41	FCC	/ACMD41/,CR,LF
2151	2281	6EA3	04		FCB	4
2152	2282	6EA4	43	MCMD9	FCC	/CMD9/,CR,LF
2153	2283	6EAA	04		FCB	4
2154	2284		*			
2155	2285	6EAB	0D	MSMENU1	FCC	CR,LF,CR,LF,/Main MENU/
2156	2286	6EB8	0D		FCC	CR,LF,/D)ir, L)oad, save B)asic,
2157						save M)code, K)ill, S)etSD, O)ption, Q)uit: /
2158	2287	6F00	04		FCB	4
2159	2288	6F01	0D	MSMENU2	FCC	CR,LF,CR,LF,/Option MENU/
2160	2289	6F11	0D		FCC	CR,LF,/1:Read Sct, 2:Disp SD
2161						data, 3:Disp MBR, 4:Disp BPB, 5:Disp FAT, 0:Return: /
2162	2290	6F5D	04		FCB	4
2163	2291	6F5E	53	MSSCT	FCC	/SctNo: /
2164	2292	6F65	04		FCB	4
2165	2293	6F66	0D	MSSD0	FCC	CR,LF,/SD Card/
2166	2294	6F6F	04		FCB	4
2167	2295	6F70	0D	MSSD1	FCC	CR,LF,/SDHC Card/
2168	2296	6F7B	04		FCB	4
2169	2297	6F7C	0D	MSINIT	FCC	CR,LF,CR,LF,/SD Card
2170						setting.../
2171	2298	6F92	04		FCB	4
2172	2299	6F93	53	MERR	FCC	/Setting error!/,CR,LF
2173	2300	6FA3	04		FCB	4
2174	2301	6FA4	49	MSERR	FCC	/Input error!/,CR,LF
2175	2302	6FB2	04		FCB	4
2176	2303	6FB3	0D	MSOFS	FCC	CR,LF,/Sector no is:/

2177	2304	6FC2	04	FCB	4
2178	2305	6FC3	0D	MSBPB FCC	CR,LF,/ BPB: /
2179	2306	6FCB	04	FCB	4
2180	2307	6FCC	0D	MSFAT FCC	CR,LF,/ FAT: /
2181	2308	6FD4	04	FCB	4
2182	2309	6FD5	0D	MSDIR FCC	CR,LF,/ DIR: /
2183	2310	6FDD	04	FCB	4
2184	2311	6FDE	0D	MSCL2 FCC	CR,LF,/ 1stCLUSTER: /
2185	2312	6FED	04	FCB	4
2186	2313	6FEE	0D	MSINFN FCC	CR,LF,CR,LF,/ FileName: /
2187	2314	6FFD	04	FCB	4
2188	2315	6FFE	0D	MSMSV FCC	CR,LF,CR,LF,/ FileName & Param:
2189	/				
2190	2316	7015	04	FCB	4
2191	2317	7016	0D	MSLDNG FCC	CR,LF,/ Loading /
2192	2318	7021	04	FCB	4
2193	2319	7022	0D	MSWTNG FCC	CR,LF,/ Writing /
2194	2320	702D	04	FCB	4
2195	2321	702E	0D	MSLDED FCC	CR,LF,/ Load end./
2196	2322	703A	04	FCB	4
2197	2323	703B	0D	MSWTED FCC	CR,LF,/ Write end./
	2324	7048	04	FCB	4
	2325	7049	0D	MSNOFL FCC	CR,LF,/ Not exist! /
	2326	7056	04	FCB	4
	2327	7057	0D	MSNXTF FCC	CR,LF,"Next page (SP:Next): "
	2328	706E	04	FCB	4
	2329	706F	3E	MSOVR FCC	/>64k/
	2330	7073	04	FCB	4
	2331	7074	0D	MSKILL FCC	CR,LF," Kill OK?(Y/N): "
	2332	7086	04	FCB	4
	2333	7087	0D	MSEXST FCC	CR,LF," Exist! Over Write?(Y/N): "
	2334	70A3	04	FCB	4
	2335	70A4	0D	MSMCOF FCC	CR,LF," -\$2000 Offset?(Y/N): "
	2336	70BC	04	FCB	4
	2337				
2338					*****
2339					
2340		END	START		

0 error(s), 0 warning(s) is detected.

Label address

ABLOP	68D9	ACMD41	6E71	AD1	6CFB	AD2	6D0C	AD3	6D16
AD4	6D1C	ADD0	6B52	ADD1	6B60	ADD2	6B68	ADD3	6B70
ADD40H	6B50	ADD42	6CF5	ADX0	6B41	ADX1	6B47	ADX2	6B4B
ADX3	6B4F	APLUSB	690D	ASC2BIN	679D	ASC411	620D	ASCMD41	6200
BASADP	0033	BASAEP	0035	BASEXP	605D	BI21	6C5C	BI22	6C62
BUSY	6AB9	CALCHC	6C28	CALCSD	6C1D	CALLOP	6959	CKRS	6AAF
CL1	6CA2	CLEND	6CB2	CLLOP	6CAD	CLOP	6C80	CLRDIR	6725
CLSTH	6048	CLSTL	604A	CMD0	6E47	CMD1	6E4D	CMD17	6E53
CMD24	6E59	CMD55	6E6B	CMD58	6E65	CMD8	6E5F	CMD9	6E77
CMPEND	65F4	CMPFN	65D9	CMPLOP	65E1	CNT	604C	CNV1	67ED
CNV2	67F9	CNVBUF	6077	CNVSUB	67E5	CR	000D	CSR0N	FBF7
CVBEND	6CEF	CVBIN	6CDD	CVERR	6CF2	CVT1	6CBB	DB2	6D8B
DB2ASC	6D7C	DBLOP	6D82	DCL0P1	6DF8	DCL0P2	6E07	DELMK	00E5
DEVN0	00BF	DIRENT	601F	DIRGET	6891	DIROFS	605B	DLOP1	6873
DPGN0	605A	DSEND	6DE5	DSL0P1	6DB2	DSL0P2	6DCD	DSPB1	6BB5
DSPB2	6BB7	DSPB3	6BC8	DSPBIN	6C48	DSPBPB	63DB	DSPBUF	6BAB
DSPC1	6474	DSPCKE	642C	DSPCLS	6DEC	DSPDIR	63FF	DSPED	6890
DSPEND	64AC	DSPFAT	63ED	DSPFL0	640B	DSPFL1	641C	DSPFL2	6436
DSPFL3	6438	DSPFL4	6447	DSPFL5	6494	DSPFL6	64A0	DSPMBR	6387
DSPSDDAT	62F3	DSPSIZ	6D97	DSPYMD	6D36	DSS1	6DA5	DSS12	
6DBB									
DSS13	6DC0	DSS14	6DCB	DSS2	6DD6	DSS21	6DD9	DSS3	6DDD
DYMD1	6D48	DYMD2	6D57	EEND	6CDC	EREND0	6CD5	EREND1	6CD7
ESC	001B	EXIST	6852	EXSTFL	6065	FILBUF	607B	FILL	6B38
FIL0P	6B3A	FILSIZ	6046	FMDAT	02FA	FNAME	604F	FNIN0	6E18
FNMIN	6E11	GET1	6392	GET2	6394	GET8CH	6C7C	GETBPB	63C2
GETD1	6BE9	GETDAT	6BE5	GETFDT	65F6	GETMBR	63A5	GR00	617A
GR2580	623A	GR2581	6251	GR2582	6253	GR410	6215	GR550	61F6
GR580	61CC	GR581	61D6	GR582	61D8	GR80	619C	GR81	61AE
GR90	6306	GR91	6312	GTBL0P	63CD	GTF1	6615	GTF2	6618
ILOP1	6276	INCHA	FBB8	INCX4	6B41	INEND	6C8E	INIT	60BF
INOUTC	6E2D	INPBUF	606F	INPFLDAT	6780	INSCT	6BFF	ISTT	
6160									
KILEND	66C0	KILFILE	666C	KILSUB	6685	KL1	66AA	KL2	66B8
LD1	6648	LD2	665C	LD3	64D2	LD41	64F7	LD5	6510
LD8	6552	LD80	6570	LD81	6574	LD9	6577	LDE0	6597
LDE1	659D	LDEND	6587	LDFILE	64AD	LDFL0	6624	LDFL1	6630

LDFL5	6654	LDFLE	666B	LDLOP	6520	LDNEND	65A1	LEND	6E46
LF	000A	LININ	6E34	LININP	D807	LINLOP	6E3C	MAIN	609B
MATCH	65EE	MCADP	605F	MCAEP	6061	MCEXP	6063	MCFLG	604D
MCMD0	6E7D	MCMD41	6E9B	MCMD55	6E93	MCMD58	6E8B	MCMD8	6E84
MCMD9	6EA4	MEMSET	65A9	MENU	60C2	MENU1	60D9	MENU2	60E2
MENU3	60EB	MENU4	60F4	MENU5	60FD	MENU6	6105	MENU7	610D
MERR	6F93	MKF1	68B0	MKF2	68B4	MKFLDT	6897	MS0	65C0
MS1	65C4	MS3	65D3	MSBPB	6FC3	MSCL2	6FDE	MSDIR	6FD5
MSERR	6FA4	MSEXST	7087	MSFAT	6FCC	MSINFN	6FEE	MSINIT	6F7C
MSKILL	7074	MSLDED	702E	MSLDNG	7016	MSLOP	65C6	MSMCOF	70A4
MSMENU1	6EAB	MSMENU2	6F01	MSMSV	6FFE	MSNOFL	7049	MSNXTTP	7057
MSOFS	6FB3	MSOVR	706F	MSSCT	6F5E	MSSD0	6F66	MSSD1	6F70
MSWTED	703B	MSWTNG	7022	NOEXST	6855	NOFEND	66CE	NOTM	65F2
OPMEN10	6144	OPMEN6	6124	OPMEN7	612C	OPMEN8	6134	OPMEN9	613C
OPMENU	6115	OUTCHA	FBBB	PBCTL	FD9F	PBDAT	FD9E	PCRLF	6C71
PSEND	6D33	PSLOP	6D28	PSTRNG	6D26	RD1	6B88	RD1SCT	6B75
RD2	6B8F	RD3	6B9F	RDSCT	6374	RESDAT	6008	RPT41	61E1
RWBUF	5E00	SC01	6171	SC2581	6232	SC551	61EE	SC581	61C4
SC80	6194	SC91	62FE	SCL0P1	632F	SCL0P2	6342	SCL0P3	6355
SCL0P4	6368	SCMD0	6169	SCMD258	6225	SCMD55	61E1	SCMD58	61B7
SCMD8	6187	SCTBPB	600A	SCTCLT	601A	SCTDIR	6012	SCTDSV	6016
SCTFAT	600E	SCTN0	6004	SCTN1	6005	SCTN2	6006	SCTN3	6007
SCTPCL	601E	SD1	6267	SDERR	614C	SDHC	6009	SDINIT	6153
SDW1	6A7B	SDW2	6A83	SETCKH	6C3F	SETCKL	6C36	SETCL	683D
SETFNM	670D	SETSCT	6C07	SETSNO	63B5	SETX	682C	SFATED	6848
SFLOP	6C9B	SKP1	62AD	SKP2	62EA	SKPDLM	6796	SNDD1	6BD0
SNDDAT	6BC9	SNDDT1	6A9E	SPC	0020	SRC0	680C	SRC00	6849
SRCDIR	6858	SRCDLM	678D	SRCFAT	6800	SRCFL0	6867	SRCFL1	686A
SRCFL5	6885	SRCL0P	6815	SRDLM1	678F	SRL0P1	67AE	SRL0P2	67C6
START	6000	STDATE	68D0	STTADR	6000	TPGFLG	604E	TRANS	6B2D
TRLOP	6B2F	TXTADR	2000	VN	6003	WFSIZ	6042	WK1	6067
WK2	6069	WK3	606B	WK4	606D	WMDY	603F	WR2	6995
WR3	69BA	WR4	69F0	WR5	69F2	WR6	69F4	WRB0	66D5
WRB1	66F3	WRBEND	670C	WRBFILE	66C1	WRDLOP	6AD7	WRDSKP	6ADC
WREND	69FF	WRF1	6B10	WRF2	6B13	WRF3	6B29	WRFL1	671A
WRFL2	6724	WRFLP2	6715	WRL0P	69A5	WRL0P1	6A44	WRM1	674D
WRMEND	677F	WRMFILE	6733	WRS0	6A25	WRS1	6A3A	WRS2	6A63
WRSUB	6A06	WRT3	694C	WRT4	6951	WRT5	6954	WRTDIR	6AC4
WRTF1	692C	WRTF2	6930	WRTF3	6971	WRTF4	6973	WRTFAT	6AF1

WRTFIL 691F WRTSD 6A64 WRTSD1 6A67 WSPFL0 6ACF WWCL0P 6A10
<!--temp-->

SDRWFM

I've uploaded all the programs I created to [OneDrive](#) . The contents include the four programs mentioned above (SDRWFM, STARTUP, RTC_SET, SDTRS), as well as the previously released RTC read/write programs (RTC_RD, RTC_WR) and a brief explanation of how to use them.

lastly

I've long wanted to implement SD card reading and writing using only 6809 assembler. When I searched online, I found that most of the solutions used the FAT library for microcontrollers like Arduino, and I was hesitant to use a microcontroller with an 8-bit retro computer.

At that time, I came across [Mr. Hishida's blog](#) from Chunichi Denko , and I decided to take the first step, thinking that if it could be done with the Z80, it should be possible with the 68 as well.

Further searching led me to a practical example ([by Chikurappe-san](#)) of controlling an SD card without using a microcontroller on a Z80 series processor, but I couldn't find anything for a 68 series processor. That was a bit disappointing, but it motivated me to try something different, and I've managed to get this far.

The fundamental requirement was to create a system for exchanging files between a Windows PC and an FM-7, so it had to be implemented using a general-purpose Windows file system without using any special saving formats. This meant that even peripheral aspects, such as timestamps, required considerable effort.

The 6809 assembler code exceeds 4KB, and it's about to clash with the DISK code from the \$71D5, so I think further functionality expansion is impossible on the FM-7. There's also the option of using hidden RAM, but since it's intended to be a utility program that runs only when needed, not a resident program, I don't think it's suitable. I think porting it to the 77AV so that it can be used with AV systems would broaden its applications, and I'd like to make that happen someday.

I mentioned this last time, but I'd like to express my gratitude once again to the blogs and other resources I used as references when creating the program. Thank you very much.

- ◎ Blog of Mr. Hishida from Chunichi Denko... SD/SDHC reading and writing techniques (with Z80)
- ◎ELM by ChaN's blog... Manual-like materials (in English) regarding SD and SDHC
- ◎FM-7 F-BASIC Analysis Manual Phase II... F-BASIC Analysis (I referred to the online version. Without this, it would have been impossible to load F-BASIC programs saved in ASCII format.)